lumina-par — Design Document

PCB-0005-A

hwde v1 pipeline

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1 Board and Run Metadata

Field	Value
board	lumina-par
part number	PCB-0005-A
workspace	`boards/PCB-0005-A_lumina-par`
phase	P8
gate status	5/6 gates pass (not passing: dfm)
state created	2026-07-28T15:00:15
state updated	2026-08-14T01:21:02

2 Overview

3 Requirements

Requirements: lumina-par (LUM-DTR-PAR-A / LUM-PAR-A)

Sources, in precedence order (later overrides earlier): 1. `brief/00-lumina-system-context.md` - shared system context. 2. `brief/03-rgbw-par-daughter-brief.md` - this board's brief. 3. `brief/01-carrier-board-brief.md` - the mating carrier, CONTEXT ONLY. 4. `brief/05-lumina-closed-decisions.md` - BINDING; supersedes the open-decision table in `00` s6 and corrects figures in `00` and `03`. 5. `brief/06-connector-icd.md` - FROZEN interface control document (ICD-01), owned by LUM-CAR-A. Hard input. Never redefined here.

Unstated low-risk items are marked `ASSUMED:` inline. Section 9 holds every unknown that changes the design or touches safety. Section 8 flags one >30 V condition that is unconditional (48 V is on the connector whether or not this board taps it) and one conditional >3 A condition.

Two figures in the briefs are dead and are NOT used anywhere below: the "˜8.5 W sustained" daughter budget in `00` s5.1 and `03` PAR-REQ-11, and the Si3402-B / AN956 "˜10 W regulated" it derives from. The carrier uses a TPS2378-class PD controller plus a 100 V buck; the re-derived budget is **8.6-9.3 W (af) / 18.7-20.0 W (at)**. PAR-REQ-11 itself still holds hard.

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0. Formal amendments (owner decision at H1, 2026-07-28)

These requirements were **changed by decision at the H1 checkpoint**, not quietly failed. The original text is quoted in full so the record shows exactly what was altered and why. Both amendments follow from the H1-Q4 decision: **same-reel sourcing, with the EEPROM fitted but shipped empty**, because no colour-measuring instrument exists for this build today.

AMD-01 - PAR-REQ-16 changes from a binning specification to a same-reel sourcing requirement > Original (brief '03' s4, PAR-REQ-16): "LED binning specified in the BOM for > all four channels. Unbinned parts across 8 fixtures will not satisfy > PAR-REQ-06."

Amended to: *All emitters for the whole build shall be procured in a single transaction from a single reel per part number, and the reel/date code shall be recorded in the build record. A chromaticity or flux bin shall be specified in the BOM wherever the vendor publishes one; where the vendor publishes none, the same-reel requirement stands in its place.*

Why it could not be met literally: LCSC/JLC does not generally expose a chromaticity or flux bin as an orderable attribute, and the selected emitter vendor publishes no bin data at all (`research/led-emitter.md` s9). A BOM line demanding a bin that cannot be ordered is unbuildable. Parts from one reel are same-batch and closely matched in practice, which is the outcome PAR-REQ-16 was written to obtain. **This is a genuine reduction in guarantee, not an equivalent substitution:** same-reel is an empirical correlation, whereas a published bin is a contractual limit. PAR-REQ-06 (consistent colour across fixtures) is therefore carried at **reduced confidence** and its verification moves to a bench comparison of the assembled fixtures.

AMD-02 - PAR-REQ-17's calibration data is deferred, not the hardware > Original (brief '03' s4, PAR-REQ-17): "Provide for per-fixture calibration: > measured per-channel scaling stored in the daughter board's ID EEPROM > (CAR-REQ-07, EEPROM option) so firmware can normalise fixtures to each other. > If the resistor-divider ID option is chosen instead, calibration has to live on > the host, and that trade must be made deliberately."

Amended to: *The board shall fit the ID divider bottom leg AND a 24C32 I2C EEPROM. The EEPROM ships **empty**: per-channel scaling coefficients are not measured or populated for the first build. The provision is retained so that calibration can be added later without a respin.*

Why: the premise that EEPROM and divider are alternatives is not what ICD-01 says - s2/s3.3 has them coexisting, the divider carrying board type and the EEPROM per-unit calibration, which is why the second dedicated ID pin was deleted. So the "deliberate trade" the original text asks for does not arise; both are fitted. What is genuinely deferred is the **data**, because there is no colorimeter or spectrometer in this project today (H1-P10). **Consequence to carry:** until the EEPROM is populated, fixture-to-fixture matching rests entirely on AMD-01's same-reel sourcing.

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1. Function

The colour and mood workhorse of the LUMINA fixture set: an RGBW par daughter board that stacks above the universal LUM-CAR-A carrier on the frozen ICD-01 expansion connector and turns four (possibly five) carrier PWM channels into four independent constant-current LED drives. It

runs continuously rather than in bursts, so its problem is not peak power but quality of dimming and consistency between the 6-8 fixtures in the room: visually stepless dimming at low output, slow hue drift over tens of seconds with no banding, and no per-fixture tint variation in a synchronised full-room wash. It carries LED drivers, thermal sensing, an independent over-temperature shutdown, the ENABLE gate, the board ID leg and the calibration store - and nothing else. It has no MCU, no network interface, no external connector of any kind, and no energy store for burst output (that is the strobe board's problem, not this one's).

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2. Interfaces

Every interface on this board is through ICD-01. There are no others. ICD-01 s9 is explicit: a daughter may not add an external connector of any kind - no barrel jack, no DMX, no second Ethernet. Ethernet on the carrier is the fixture's only external connection, and that is what makes the non-isolated PoE topology compliant.

2.1 J3 - POWER block, 2x7, daughter side

- Part: **CONNFLY DS1023-2*7SF11** socket (14 pos, 2.54 mm, 600 V, 3 A/contact, 8.5 mm body). Second sources on the same land pattern: HanElectricity 2541FV, Boomele 2.54-2*7P, HCTL HC-PZ254. JLC Extended - there is no JLC Basic part in any board-to-board family, so "prefer Basic" is unachievable here and is not a selection error.
- **Reverse-mounted on the daughter's BOTTOM side, facing down** (ICD s7.3).
- Pin map (ICD s3.1), authoritative, do not re-derive: 1/3/5 `+48V_SW`; 2/4/6/7/8/10/13 `GND`; 9/11 `+12V`; 12/14 `+3V3`. Position 1 silkscreen-marked with a triangle.
- Rail order along the connector is 48 -> GND -> 12 -> 3.3 so that no single-position mis-seat can put a higher rail on a lower rail's pin.

> **P8 AMENDMENT 2026-08-08 - THE MAPS ABOVE ARE ICD* POSITION* NUMBERS, NOT > THIS BOARD'S PAD NUMBERS. On a bench meter they differ.** J3/J4 are > reverse-mounted on B.Cu facing down (s7.3 above), and flipping a 2-row > connector to the back **swaps its rows**: carrier position `2k+1` physically > mates this board's pad `2k+2`. The schematic pin assignment was mirrored to > compensate - verified pin-by-pin against the carrier's own netlist, 0 of 38 > contacts crossed - so the MATED pair is correct. But it means: > > * **ICD position 1 = this board's pad 2.** Probing "pin 1" by the map above > on the daughter reads `GND`, not `+48V_SW`; probing pad 1 reads 57 V while > the map says GND. Both mistakes are available and one of them is dangerous. > * The silkscreen triangle added at P8 marks the pad that mates **carrier > position 1** - i.e. it marks pad 2, deliberately. It is a MATED-VIEW mark. > A stale, unprintable circle from the footprint still sits on pad 1; ignore it. > > Before this was found, both boards transcribed the ICD map literally by pin > number, which crossed 12 of 14 J3 contacts and **shorted +48V_SW to GND at six > of them**. That is why this note exists. When probing this board, use the pad > numbers in `boards/lumina-par/work/board.net`, never the position map above.

2.2 J4 - SIGNAL block, 2x12, daughter side

- Part: **CONNFLY DS1023-2*12SF11** socket (24 pos, 600 V, 3 A/contact, 8.5 mm body), same mounting rules and same second-source situation.
- Pin map (ICD s3.2), authoritative: `PWM0..7` at positions 1,2,5,6,7,8,11,12; `GND` at 3,4,9,10,13; `DSPI_SCK` 14, `MOSI` 15, `MISO` 16, `CSn` 17; `I2C_SCL` 18, `SDA` 19; `ADC0` 20, `ADC1` 21; `ID_ADC` 22; `ENABLE` 23; `FAULT` 24. **No 48 V exists anywhere on this connector.**

2.3 Signal contract (ICD s3.3) - what this board must honour

Signal	Requirement on this board
‘PWM0..7‘	3.3 V push-pull, **13-bit at 9.766 kHz default** (CAR-REQ-12). PWM0-3 = LEDC timer 0, PWM4-7 = timer 1; channels on a timer share frequency AND resolution. ‘ASSUMED:‘ R,G,B,W map to PWM0-3 (timer 0); a 5th channel (question 4) takes PWM4 on timer 1, programmed identically. LEDC timers 2 and 3 are unallocated and available on request - that is the only sanctioned route to a different PWM frequency for this daughter, and it needs the carrier owner’s agreement, not a unilateral change.
‘I2C_SCL/SDA‘	400 kHz open drain. **Pull-ups are on the carrier (4.7 k). This board must NOT fit its own.** The daughter owns the whole address space; the carrier reserves nothing - so EEPROM and any digital temp sensor must be allocated non-colliding addresses here.
‘ADC0‘, ‘ADC1‘	daughter -> carrier, 0-3.3 V, **source impedance <= 10 kohm**. This is the PAR-REQ-12 thermistor path; an NTC divider must be sized to meet the 10 kohm ceiling at every temperature of interest.
‘ID_ADC‘	The carrier fits the TOP leg (10 k to +3V3); **this board fits the BOTTOM leg to GND**. The code value is allocated by the carrier owner, not chosen here - see question 15.
‘ENABLE‘	carrier -> daughter, push-pull, **active HIGH**. This board must fit its own **100 kohm pull-down**, gate **every** output stage with it (driver EN pins, any gate driver, any charge path), and **never latch it locally**. A carrier PWM pin can glitch ~60 us at power-up; ENABLE is what makes that a no-op.
‘FAULT‘	daughter -> carrier, **open drain, active low**, 10 k pull-up on the carrier, wire-OR’d with the carrier’s eFuse fault. **Never drive it high.**
‘DSPI_*‘	Available, <= 26 MHz, mode 0, shared bus with one CS. Not expected to be needed by a 4-channel PWM design; a daughter needing more devices decodes locally.

2.4 LED interface Emitters are the board’s real load and their mounting is **not settled** - on-board emitters versus an off-board module on a separate heatsink with an internal wiring harness is question 5, and it interacts with ICD s9’s external-connector ban (an internal harness inside the enclosure is not an external connector, but this needs confirming, not assuming). Optics (PAR-REQ-14 wide wash beam from a 2.5 m ceiling, PAR-REQ-15 diffusion sufficient that R/G/B/W mix before reaching a surface) may or may not be in this run’s scope - question 8.

2.5 Status / test points No requirement stated. `ASSUMED:` no user-visible indicators on this board - CAR-REQ-09 status indication lives on the carrier. Any test point fitted must carry the ICD s9 bench-hazard warning: the entire fixture floats at PoE potential, and an earthed scope probe or non-isolated USB-UART adapter does not merely risk damage, it **breaks PD signature detection outright** (detection currents are a few hundred microamps).

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3. Power

3.1 The envelope (binding)

	802.3af (first build)	802.3at (upgrade path)
Sustained power to this daughter, all rails combined	**8.6-9.3 W**	**18.7-20.0 W**

D-01 is closed: the carrier’s PD front end, converter, magnetics and thermal design are sized for **Type 2 (at)**, but classification is programmed to **Type 1 (af)** for the first build via a class resistor.

The upgrade is a resistor change plus a PoE+ switch, **no respin**. D-01 says explicitly: *do not allow any other component to become the part that pins the design to Type 1.* On this board that clause lands on the LED drivers, the emitters and the heatsink - see question 2, which is the highest-consequence open item in this document.

3.2 Per-rail ceilings (ICD s6.2) - individual ceilings, the TOTAL binds

Rail	Sustained af	Sustained at	Peak (ms, local bulk)	Hardware fault ceiling
`+48V_SW`	0.25 A (12.0 W)	0.50 A (24.0 W)	1.0 A	1.0 A eFuse, **latch off**
`+12V`	0.75 A (9.0 W)	1.25 A (15.0 W)	2.0 A	2.0 A converter OCP
`+3V3`	0.25 A (0.83 W)	0.25 A	0.50 A	1.0 A converter, limit >= 1.3 A

Derived, and load-bearing: the +12V rail alone cannot cover this board's envelope at either class. At af it needs 0.72-0.78 A against a 0.75 A ceiling (right at the edge); at at it needs 1.56-1.67 A against a **1.25 A** ceiling, which it misses by 3.7-5.0 W. ICD s6.3 is unambiguous: *anything above 1.25 A at the at operating point must be taken on `+48V_SW`*, and the 1.25 A ceiling is a thermal limit on the carrier's 48->12 converter in a sealed box, not a current rating that can be argued with. Taking power on `+48V_SW` instead of `+12V` is also worth **0.67 W (af) / 1.30 W (at)** of extra delivered power because it skips a conversion.

`ASSUMED:` for an **af-sized** board, LED power is taken from `+12V`, which is D-02's stated purpose for that rail (so 6-8 fixtures do not each duplicate a >= 60 V converter). If question 2 answers "size for at", that assumption is void and P1 must take at least part of the load from `+48V_SW`, inheriting the bleed path, the 0.60 mm creepage regime, 100 V capacitors, 0805-or-larger resistors across the 48 V domain, and the inrush obligation below.

3.3 Hard rules inherited from the ICD

- **Hardware backstop for PAR-REQ-11.** The clamp is a firmware rule, but firmware fails. With ENABLE asserted and every PWM stuck at 100 % (hung MCU; the 2 s watchdog fade cannot help if the MCU is the thing that hung), the board's total draw must remain **below the carrier's hardware fault ceilings** - 12 V OCP 2.0 A, 48 V eFuse 1.0 A latch-off, PSE overload timer ~50-75 ms. A board that can only meet the budget when firmware is healthy will brown out or latch off the whole fixture. This sizes the per-channel drive current and is not optional.

- **`+48V_SW` is dead at power-up and stays dead for hundreds of ms.** The carrier's load switch is a compliance part (802.3 caps PD port capacitance at ~180 uF); it closes only after firmware asserts ENABLE. Design for it.

- **No first-mate/last-mate sequencing exists.** The board must tolerate 48 V arriving before or after 3.3 V, in either order.

- **No path may energise anything on this board from `+12V` or `+3V3` while `+48V_SW` is off**, or the 802.3 inrush compliance is defeated behind the switch's back.

- **Inrush is this board's responsibility** (CAR-REQ-14). Size the limiter against the **PD's 1.0 A operating current limit**, not the connector's 5.4 A rating; sizing against the connector is the classic way to trip the PD's 800 us foldback deglitch and brown out the fixture.

- **If any 48 V net is tapped: a bleed path is mandatory** (CAR-REQ-17 / STR-REQ-10). The carrier deliberately fits no series diode on `+48V_SW`, so the daughter's bleed is not stranded above the carrier's 100 k.

- **0.60 mm outer-layer copper clearance around every 48 V net, board-wide, is NOT inherited automatically** - this board's DRC must be set up for it at P5 (IPC-2221B B2, 51-100 V band, 57 V worst case). The 0.13 mm coated column is not claimable: LPI soldermask is not a qualified conformal coating and `check_creepage.py` implements only the uncoated columns, so a layout designed to 0.13 mm fails P8 with no waiver mechanism.

3.4 LED-side budget (ESTIMATES, clearly marked as such) GUESS, for question sharpening only, to be replaced by a real power table at the P1/P2 design gate (`03` review gate 1):

- Daughter housekeeping (driver quiescent, ID/EEPROM, sense): **~0.3-0.5 W**.
- Switching CC driver efficiency: **~88-92 %**.
- Therefore electrical power reaching the emitters: **~7.6-8.5 W (af) / ~16.5-18.4 W (at)**.
- A typical 4-in-1 RGBW multi-die emitter (R AlInGaP ~2.1-2.9 V, G/B InGaN ~3.0-3.6 V, W ~2.9-3.4 V) draws roughly **4.0-4.7 W at 350 mA/die** and **8.4-9.8 W at 700 mA/die** with all four dies on. So **one** such emitter at ~600-700 mA/die consumes the entire af envelope by itself; the at envelope buys roughly double.
- Note for P1, not a decision here: PAR-REQ-10 forbids burning the R-vs-GBW forward-voltage spread in a shared linear element, and a 12 V rail against a 2.1-3.6 V single die is a 3.3-5.7x step-down. That combination rules out shared linear current control and points at either per-channel switching regulators or multi-die series strings per channel. It also interacts with question 3 (package choice).

No battery, no charging, no mains anywhere in this system.

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4. Environment

- Deployment: indoor, basement/garage room ~5 m x 7 m x 2.5 m, 8-12 fixtures, ceiling-mounted at ~2.5 m. `ASSUMED:` ceiling mount, beam pointing into the room; no vibration or shock requirement; no ingress rating (question 6 decides sealed vs vented, which is a thermal question, not an IP one).
- `ASSUMED:` external ambient 0-40 C. Not stated anywhere in the briefs.
- **Internal enclosure air reaches 56 C (af) / 69 C (at)** per ICD s7.6. This is the ambient the emitters and drivers actually see, not room temperature.
- **Continuous duty with no relief.** Unlike the strobe, this board runs hot all the time (`03` review gate 4, and H1-Q5 restates it). Every thermal number must be a steady-state number.
- Heat this board must reject (ESTIMATE: LEDs convert ~20-35 % of electrical input to light, the rest to heat): **~6.5-8.4 W (af) / ~14-17 W (at)**, plus the carrier's own 2.4 W (af) / 3.7 W (at) in the same box.
- **This is the riskiest number in the document.** With internal air at 69 C and a red-die junction that wants to stay near 85-100 C for colour and lifetime stability, the junction-to-internal-air thermal budget at the at operating point is roughly **1.0-1.1 C/W**, and at af roughly 2.3 C/W. A sealed plastic box has no convective path out. Questions 6 and 7 exist because the at case very likely does not close without either ventilation or an external heat path, and H1-Q5 constrains both (non-conductive enclosure, heatsink NOT user-accessible).
- Connector contacts are rated -40 to +105 C, so they are not the limit.

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5. Size & mounting

All of the following is the **common LUMINA footprint** from ICD s7.1, which every board inherits. It is proposed, not yet confirmed with the human - MECH-02 requires the outline to be closed at H1 **before P5**, because `board_init.py` binds the outline permanently and ai-ee has no outline-shrink step. See question 12.

```
| Item | Value |
|---|---|
| Outline | **100.0 x 80.0 mm** |
| Corner radius | **3.0 mm**, all four corners |
| Mounting holes | **4x M3 (3.2 mm) at 5 mm inset** (a 90 x 70 mm rectangle)
**plus a 5th M3 at (46, 74)** |
| Thickness | 1.6 mm |
| Coordinate origin | board top-left, x right, y down |
| Mated board-to-board height | **11.0 mm** hard-seated |
| Standoffs | 5x M3 female-female, 11.0 mm |
```

5.1 The RJ45 notch - hard requirement from P2 onward A 30 x 26 mm notch in the TOP edge, region (6, 0) - (36, 26). The carrier's board-edge magjack is ˜15 mm tall against an 11.0 mm stack, so it protrudes ˜4 mm above this board's underside. The outline rectangle, corner radius and 5-hole pattern are unchanged; only this local relief differs. It is also the **primary anti-180-degree interlock** (CAR-REQ-16): rotated, the notch lands at the bottom edge and the board presents solid material over the jack, so the boards cannot be forced flat. That is a mechanical stop, not a warning.

5.2 Exclusion zones (ICD s7.6) - all four apply to this board

```
| Zone | Region | Rule |
|---|---|---|
| RJ45 relief | (6, 0) - (36, 26) | cut away entirely (s5.1) |
| DC-DC hot zone | (2, 46) - (36, 68) | **no LED drivers and no aluminium
electrolytics** - the carrier's 48->12 converter dissipates up to 1.25 W
directly below, and in a stacked mezzanine this board's parts sit vertically
over it |
| Antenna column | (88, 25) - (100, 55) | **no copper on any layer, no metal
component** - the carrier's ESP32-S3 PCB antenna is 11 mm below and Wi-Fi is a
supported control path (H1-Q8) |
| Recovery header | (76, 0) - (98, 20) | keep clear enough for a 6-way jumper
lead with this board fitted, or accept board removal for firmware recovery |
```

5.3 Connector positions - PROVISIONAL, and this board is blocked on them ICD s7.2 gives J3 body (14, 68)-(34, 78) with position 1 at (15.3, 69.3), H5 at (46, 74), J4 body (56, 68)-(88, 78) with position 1 at (57.3, 69.3). **ICD s7.2 is the one section not frozen at H1**: the carrier owner must compare placed positions after its own P6, correct them, and re-issue. Daughters are explicitly blocked on that re-issue. See question 16.

Mating geometry to verify at P4/P6, not to re-derive: this board's sockets are **bottom-side, facing down**, and mate with a top-side header on the carrier. Plan-view (x, y) coordinates are shared between the two boards, but pin numbering mirrors across the row on a bottom-side footprint - pin 1 alignment must be checked in the mated view, not assumed from the footprint.

5.4 Pipeline facts that constrain P5 (verified in this repo, not assumed)

- `board_init.py --mounting-holes N` places holes at **inset = margin / 2** (confirmed in the script). The ICD's 5 mm inset therefore requires **'–margin 10'**, not the default 6. The corner radius is separately clamped in the worker; read the report's `corner_radius` field and `worker_notes` rather than assuming the requested 3.0 mm was honoured.

- **'board_init.py' has NO notch/cutout option** (`--outline` takes only `auto` or `WxH`), and `kc.py` exposes no outline-editing subcommand. The mandatory 30 x 26 mm relief must therefore be produced by a direct Edge.Cuts edit of the `.kicad_pcb` after `board_init`, with every downstream consumer (`place_*`, `planes_gen`, `dfm_check`, `fab_export`) re-verified against the modified outline. **Flagged as an implementation risk for P5**, not a requirements question - the notch itself is non-negotiable.
- Placement keepouts ARE supported: `constraints.json` accepts `placement.keepouts: [{"rect": [x1,y1,x2,y2], "reason": ...}]` and the annealer treats them as fixed obstacles. All four ICD s7.6 zones plus the notch region must be declared there.

5.5 Height Board-to-board is 11.0 mm. **Total height above this board is unknown** and depends on the emitter/heatsink/optic stack (questions 5-8). Not stated anywhere in the briefs.

—

6. Quantity & budget

- **Quantity: 6-8** par fixtures out of 8-12 total (`03` s1). Exact count and spares not stated - question 14.
- System budget: **$500-1000 for fixtures + switch + enclosures** (`00` s1). ICD s1.1 rejects a Samtec pair at $15.30 "over half the $30/board target", so a **$30/board** figure exists in the carrier run - but no unit cost target has been stated for *this* board. With 8-12 fixtures, a managed PoE+ switch (note: D-01's upgrade path implies PoE+, which is materially more expensive than the PoE switch the original budget assumed), enclosures, carriers, strobes and pars all inside $500-1000, this is tight. Question 14.

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7. Assembly

- `ASSUMED:` JLCPCB fabrication and PCBA, consistent with every other board in this repo. Not stated in the LUMINA briefs.
- **Both ICD connectors are 2.54 mm THT sockets that must be reverse-mounted on the bottom side.** That is not a normal JLC PCBA process, so the realistic routes are hand-soldering them post-PCBA or finding a bottom-side SMD equivalent on the same land pattern (ICD s7.3 permits "a bottom-side SMD equivalent"). Question 13.
- High-power emitters, if on-board, need a large thermal pad and a via farm on 1.6 mm FR4 - or an MCPCB, which the common footprint does not provide. Interacts with questions 5 and 13.
- Sidedness: bottom side already carries the two sockets, so the board is double-sided by construction. Question 13 confirms the process.

—

8. Compliance / safety flags

```
| Flag | Status |
|---|---|
| Mains voltage | **No.** PoE only.  |
| Batteries | **No.** No battery, no charging anywhere in the system.  |
| Motors | **No.** |
| **> 30 V** | **YES, unconditionally.** '+48V_SW' is present on J3 whether or
not this board taps it.  PD rail 37-57 V nominal, **57 V worst case** (IEEE
```

802.3 PSE maximum). Consequences: 0.60 mm outer-layer clearance around every 48 V net board-wide; 100 V capacitors (63 V is not enough at 57 V once ceramic DC-bias derating applies); any resistor across the 48 V domain must be 0805 or larger, or split in series; the clearance applies through the board (inner 0.10 mm / outer 0.60 mm under any 48 V antipad); mandatory bleed path if tapped. **Insulation class is functional only** - 57 V is below the IEC 62368-1 ES1 limit of 60 V DC, so no personnel safeguard and no safety-mandated creepage applies, and an unearthed PD needs **no MOV-to-earth surge network** (do not copy one from a reference design). |
High current (> 3 A)	**Conditional - architecture must report and re-flag.** No connector rail exceeds 1.25 A. But LED-side per-channel current is unbounded by the ICD: a single low-Vf red channel sized near the whole envelope (e.g. 8 W into a ~2.4 V string) would be ~3.3 A. P1/P2 must publish per-channel drive currents; if any exceeds 3 A this flag becomes unconditional.
RF transmit	**No radio on this board** - but the carrier's ESP32-S3 radio is a *supported control path* (H1-Q8), so the (88, 25)-(100, 55) antenna column is a hard keepout (no copper on any layer, no metal component) and switching LED drivers 11 mm above a 2.4 GHz PCB antenna are a desense risk that layout sign-off must address.
Floating PoE potential	**Applies to everything this board touches.** The entire fixture is non-isolated and floats at PoE potential: this board, its drivers, its LED wiring, and any heatsink the LED module sits on. 802.3 compliance is achieved *only* by there being no accessible external conductor. This is ICD s9 / ‘decisions.md‘ OPEN-C and it is **unresolved** - questions 5 and 7.
Continuous hot surface	High-power LEDs at 8.6-20 W continuous inside a non-conductive enclosure whose internal air already reaches 56-69 C. Enclosure touch temperature and plastic deformation are real; H1-Q5 requires the heatsink to be non-user-accessible.
Photobiological (IEC 62471)	A high-power RGBW source with a wash optic is plausibly Risk Group 1-2 at close range (blue-light hazard). Not a certification requirement for a private installation, but worth a note in the design doc rather than discovering it by looking into a fixture.
Certification	‘ASSUMED:‘ private installation, own use - no CE/FCC/UL conformity assessment path required.

Nothing in section 8 is guessed. The >30 V condition and the floating-PoE condition are stated facts from the ICD; the >3 A condition is explicitly left open for the architecture to answer.

—

9. Open questions

Questions 1-4 are the four the orchestrator already queued, sharpened with the numbers above. Questions 5-16 are everything else this design genuinely needs. Each offers a recommendation. Questions **2, 6, 7 and 12** are the ones that change the board most, and 12 and 16 must be answered before P5 regardless.

1. Total-power clamp policy (PAR-REQ-11). Four channels cannot all run at 100 % - the sum of four channels at full drive exceeds the 8.6-9.3 W (af) envelope by roughly 30-60 % on any sane sizing. When the host asks for more than the fixture can deliver, what should the fixture protect? (a) **Preserve hue**: scale all four channels by the same factor. The colour is exactly right; the fixture is simply dimmer. White (all four on) is the worst case and gets dimmest. (b) **Preserve brightness**: hold the total at the cap and reallocate between channels. Brightness is stable; the hue drifts toward whichever channel is cheapest in watts, and drifts *differently* per fixture, which is exactly the artifact PAR-REQ-06 exists to prevent. (c) Hybrid: preserve hue, let the white channel absorb the deficit - this desaturates at high intensity, the "visible artifact, not graceful degradation" the brief warns about. **RECOMMEND (a).** Uniform hue-preserving scaling. Colour consistency across 6-8 fixtures washing one wall is this board's reason to exist; a uniform brightness loss is

invisible, a hue split across fixtures is not. Note that whichever answer is chosen, the **hardware backstop in s3.3 is still required**

- the board must be electrically incapable of exceeding the carrier's fault

ceilings with every PWM stuck at 100 %.

2. Size the LED engine for af or at? D-01 kept the *carrier* at-capable and af-classified, and told us not to let any component pin the design to Type 1. Does that clause bind this board's LED engine too? (a) **Size for af (8.6-9.3 W)**: smaller emitter, smaller heatsink, cheaper, thermally plausible in a plastic box - but a PoE+ upgrade then buys nothing on the par because the emitters and heatsink cannot use the extra ~10 W, and getting it later is a respin of this board. (b) **Size for at (18.7-20.0 W), run clamped at af**: the upgrade is a firmware constant. But it roughly doubles heatsink area and emitter cost per fixture, and - derived above - **it forces at least 3.7-5.0 W to be taken from `+48V_SW`** because the +12V rail tops out at 15.0 W, which means a >= 60 V input stage on this board after all, the thing D-02's 12 V rail was created to avoid on 6-8 fixtures. **RECOMMEND (a) for the first build, with a written note in the design doc that the par does not follow the carrier's at upgrade.** The par is a sustained wash, not the fixture that needs the headroom; the strobe is. Spending the at budget on the par costs money and heatsink on all 6-8 units to buy brightness the profiles ask of the strobe. But this is a lighting-design call, not an engineering one - if a 9 W par reads as underwhelming, (a) is a respin.

3. LED package: four discrete emitters or an integrated RGBW multi-die? (a) **Integrated 4-in-1** (single package, four dies on one thermal slug): mixes better optically and largely solves PAR-REQ-15 fringing; one thermal interface; one binning transaction. All four dies share one junction temperature, so the hottest channel heats the others, and red (AlInGaP) is both the most thermally sensitive and the one that gets cooked. (b) **Four discrete emitters**: independent thermal management, cheaper and easier sourcing, per-channel binning - but four separate optical sources need real diffusion to avoid visible R/G/B/W shadow fringing on a wall (PAR-REQ-15), and four packages spread over the board is four thermal paths to solve. **RECOMMEND (a) integrated 4-in-1**, on the grounds that PAR-REQ-15 is a hard optical requirement and mixing is the one thing an integrated package gives for free. Note the interaction with s3.4: a 12 V rail against a single 2.1-3.6 V die is a 3.3-5.7x step-down, which pushes toward per-channel switching drivers either way.

4. A fifth channel (amber)? PWM budget is free - the connector carries 8 channels and the par uses 4, and PWM4 sits on LEDC timer 1 which can be programmed to the same 13-bit/9.766 kHz. So the cost is not connector budget, it is: one more driver, and a fifth channel drawing from the **same** 8.6-9.3 W envelope, making the question-1 clamp bite harder in every mixed colour. P3 French melodic leans on gold/amber and RGBW amber is a mix rather than a true emitter. Caveat worth weighing: amber AlInGaP loses output at roughly -0.5 to -1 %/C of junction temperature, roughly twice as badly as the InGaN channels, in a box whose internal air is already at 56-69 C continuously - so a calibrated amber will drift visibly as the fixture warms up unless it is temperature-compensated in firmware. **RECOMMEND: no fifth channel on rev A.** Add it only if question 2 answers "size for at" (where there is power to spare). At af it dilutes the envelope across five channels and adds the most thermally unstable emitter in the catalogue to a board that already has a thermal problem.

5. LED mounting and wiring: on-board emitters, or an off-board LED module on a separate heatsink? (a) **On-board**: emitters on this PCB's top side, heat through thermal vias into a heatsink bolted to the board. No wiring at all, so ICD s9 is trivially satisfied - but 1.6 mm FR4 is a poor thermal path, the common footprint gives no MCPCB, and the emitters' position is then dictated by the connector/keepout geometry rather than by the optics. (b) **Off-board module** on its own MCPCB/heatsink, connected by an internal wiring harness (5-9 conductors). Best thermal and optical freedom. **ICD s9 bans a daughter from adding "an external connector of any kind" - a harness that never leaves the enclosure is not an external connector on any reasonable reading, but this needs confirming, not assuming**, because if it is read strictly then option (b) is illegal

and option (a) is the only route. **RECOMMEND (b), with explicit confirmation that an internal wire-to-board connector (e.g. a JST-family header) is permitted** on the grounds that ICD s9's intent is "nothing penetrates the enclosure wall", and s9 itself already anticipates the off-board case ("if the LED module is on a separate heatsink (Q4a default), that heatsink and its wiring are at PoE potential too"). If the answer is a strict no, please say so plainly - it forces (a) and changes the thermal design completely.

6. Is the enclosure sealed, or vented? ICD s7.6 says "sealed box" with internal air at 56 C (af) / 69 C (at); H1-Q5 says plastic, non-conductive, with the heatsink not user-accessible. No ingress requirement is stated anywhere. This matters more than any other mechanical question: from s4, a *sealed* plastic box gives a junction-to-internal-air budget of ~2.3 C/W at af and ~1.0 C/W at at, and the at case very likely does not close. (a) Fully sealed (no openings at all). (b) **Vented** - louvres or a labyrinth sized so no finger or 4 mm probe reaches live parts or the heatsink, preserving H1-Q5's non-accessibility while allowing convection. (c) Sealed but with the heatsink forming part of the enclosure wall (see question 7). **RECOMMEND (b) vented with a finger-guard geometry.** It is an indoor basement/garage installation with no stated dust or moisture requirement, and convection is close to free. Sealing costs real money in heatsink area and caps the fixture's output.

7. Is an external or enclosure-integrated heat path allowed? ICD s9 and `decisions.md` OPEN-C leave this open and it is the sharp end of the non-isolated topology: any heatsink bonded to this board's ground **floats at PoE potential**, and "if the heatsink is touchable, metal, or shares a mount with anything earthed, the non-isolated topology is non-conformant." (a) Heatsink entirely internal, no metal reaches the outside. Thermally worst, compliance-simplest. (b) Heatsink forms part of the enclosure but is shrouded behind a plastic guard - not touchable, and not sharing a mount with anything earthed. (c) Exposed metal heatsink - **would break the compliance argument** unless the whole fixture is re-isolated, which is out of scope. **RECOMMEND (b)**, plus a hard rule that the ceiling mount is non-conductive and bonded to nothing. Also please confirm: should the LED module's metal substrate be electrically tied to board GND, or thermally coupled but electrically floating? (Recommend: floating from board GND, on a dielectric, so a single insulation failure does not put 48 V-referenced potential on the heatsink.)

8. Does this run own the optics? PAR-REQ-14 (wide wash beam from 2.5 m, overlapping coverage from 6-8 fixtures) and PAR-REQ-15 (diffusion sufficient that the colours mix before reaching a surface) are stated as requirements on this board, but a PCB run can only choose the emitter and provide mechanical provision for a lens or diffuser. (a) This run selects the emitter and provides lens-holder mounting provision; the diffuser/lens is an enclosure item chosen later. (b) This run owns the full optical chain including the lens/diffuser part numbers and beam-angle verification. **RECOMMEND (a)** - a PCB pipeline cannot verify a beam angle, and PAR-REQ-14 / 15 should be carried as enclosure requirements in the design doc so they are not silently lost.

9. What does "5-10 % of full output" mean in PAR-REQ-01? This changes the required driver PWM bandwidth by roughly 40x and it is the single most likely reason this board fails its own review gate 2. At 13-bit / 9.766 kHz the period is 102.4 us and one LSB is **12.5 ns** - no constant-current driver on the market resolves that, so the real dimming floor is the driver's PWM settling time, not the PWM word length. (a) **5-10 % of PWM duty**: on-time 5.1-10.2 us. Most switching CC drivers manage this at 10 kHz with some amplitude error. (b) **5-10 % of perceived brightness** (i.e. after gamma ~2.2): duty is 0.137-0.63 %, on-time **141-646 ns**, which is at or below the settling time of essentially every CC driver and is where "visible stair-stepping during slow fades" is actually born. `CORRECTED 2026-07-28 (orchestrator, verified):` this line previously read "on-time 1.4-6.1 us", a 10x error. The duties were right; the on-times were not. 0.05^2.2 x 102.4 us = 140.6 ns and 0.10^2.2 x 102.4 us = 646.1 ns. The correction is load-bearing: it moves (b) from "several drivers manage it" to "no surveyed CC driver manages it with PWM alone" - the best part found in P1 (TPS92515HV) specs a 0.2 % minimum duty at 10 kHz, i.e. a 200 ns floor, which is above the 141 ns that 5 % perceived requires. See `research/led-driver.md` and `research/spec-dimming.md`. **RECOMMEND: assume (b)**, because that is what a lighting person means by "dim to 5 %", and it is the harder requirement - designing for (b) satisfies (a) automatically. Please confirm, because

it materially narrows the driver shortlist and is the requirement PAR-REQ-09 is really about.

10. Calibration path (PAR-REQ-17) - and who does the measuring? The premise that this is EEPROM *versus* the resistor divider is **not what the ICD says**. ICD s2 and s3.3: the carrier fits the top leg of the `ID_ADC` divider and the daughter fits the bottom leg **and** an I2C EEPROM may ride the shared I2C bus; a second dedicated ID pin was deleted precisely because the EEPROM does not need one. So the two mechanisms coexist: **the divider says what board type this is, the EEPROM stores per-unit calibration.** A 24C02/24C32-class part is a few cents and one address allocation (this board owns the whole I2C address space - the carrier reserves nothing - so EEPROM and any digital temp sensor must be allocated non-colliding addresses). **RECOMMEND: fit both.** The real open question is downstream: PAR-REQ-06 requires measured per-channel scaling, so **what instrument measures 6-8 fixtures?** (colorimeter / spectrometer / phone-camera comparison / none). If the answer is "nothing", the EEPROM ships empty and colour matching falls back entirely on binning (question 11), which changes the sourcing strategy.

11. LED binning versus the assembly route. PAR-REQ-16 requires binning specified in the BOM for all four channels; "unbinned parts across 8 fixtures will not satisfy PAR-REQ-06". But JLC's catalogue does not generally let you specify a chromaticity/flux bin, so binned high-power emitters usually mean hand-sourcing from a distributor and hand-placing them. (a) Hand-source binned emitters and place them by hand (or as a separate LED sub-assembly), everything else JLC PCBA. (b) Accept whatever bin JLC ships and rely entirely on per-fixture calibration (question 10) to match the fixtures. (c) Both: loose binning plus calibration. **RECOMMEND (c)** - buy the tightest bin available from a distributor for one batch of 6-8 fixtures (they will then be same-reel and closely matched anyway) **and** fit the calibration EEPROM. Note (b) alone is only viable if question 10 has a real measurement instrument behind it.

12. Confirm the board outline - MECH-02, needed before P5. The ICD proposes the common footprint: **100.0 x 80.0 mm, 3.0 mm corner radius, 1.6 mm thick, 4x M3 at 5 mm inset plus a 5th M3 at (46, 74), and the mandatory 30 x 26 mm notch at (6, 0)-(36, 26)**. ai-ee has** no outline-shrink step** - whatever is passed to `board_init` at P5 is permanent, so this must be confirmed, not inherited silently. **RECOMMEND: confirm 100.0 x 80.0 mm as-is.** The carrier and daughters share one enclosure and mate through the connector, so the outlines are not independent, and after the four exclusion zones and the notch this board has roughly 60 x 74 mm of genuinely usable area for drivers plus whatever the LED scheme needs. Please confirm explicitly, including whether the enclosure has any dimension this footprint must fit inside that has not been stated.

13. Assembly method and process. Not stated anywhere in the LUMINA briefs. (a) JLCPCB fabrication + PCBA, JLC Basic parts preferred where they exist (note: the ICD connectors are Extended, unavoidably - there is no Basic board-to-board part). (b) Hand assembly. Two sub-questions that need answering with it: **are the two 2.54 mm sockets hand-soldered post-PCBA** (they are THT and must be reverse-mounted on the bottom side, which is not a normal JLC process), and **are high-power emitters hand-placed** (question 11)? **RECOMMEND: JLC PCBA for everything it can do, with the two sockets and the emitters hand-soldered afterward.** 6-8 boards is a quantity where two hand operations per board is a couple of hours total, and it removes both awkward processes from the critical path.

14. Build quantity and unit cost target for THIS board. The brief says 6-8 pars of 8-12 fixtures; no per-board cost target is stated (the $30/board figure in ICD s1.1 belongs to the carrier). The $500-1000 system budget also predates D-01's at-capable power stage, and the upgrade path implies a **PoE+** switch, which is materially more expensive than the PoE switch that budget assumed. **RECOMMEND: confirm 8 boards built (6-8 deployed plus spares) and set an explicit BOM target for this board.** A defensible starting number is **$25-35/board** excluding the LED module and heatsink, with the emitter and heatsink budgeted separately because question 2 can double them.

15. 'ID_ADC' code allocation. ICD s3.3: "Board-type codes are allocated by the carrier owner, not chosen by daughters." This board needs its allocated bottom-leg resistor value (and the

resulting ADC code) before P4 can place the part. **RECOMMEND: request the LUM-PAR-A code from the carrier owner now**; if none exists yet, ask for one to be allocated rather than picking a value here - picking one is exactly the silent divergence ICD-01's preamble forbids.

16. Sequencing against ICD s7.2 (the one un-frozen section). The J3/J4 coordinates in s7.2 are provisional until the carrier owner confirms them after its own P6 and re-issues the ICD, and daughters are explicitly blocked on that. Everything else in the ICD is frozen. (a) Wait for the re-issued ICD before starting this run. (b) **Proceed through P1-P4 (architecture and schematic, which do not depend on mm coordinates) and hold at P5**, where the outline and connector placement are bound permanently. **RECOMMEND (b)**, with the s7.2 coordinates treated as unconfirmed everywhere they appear, and an explicit re-check against the re-issued ICD before `board_init` is run. Please confirm this is acceptable, and if the carrier's P6 has already completed, point this run at the re-issued ICD instead of the DRAFT-A copy in `brief/`.

4 Architecture

Decisions of record (state.json)

Phase	What	Why
P0	D-01/D-02/D-03 inherited as closed; par power budget 8.6-9.3 W af / 18.7-20.0 W at	05-lumina-closed-decisions.md is binding and supersedes the ~8.5 W figure in briefs 00/03 (derived from Si3402-B, a part this system does not use)
P0	ICD-01 (brief/06-connector-icd.md) treated as a frozen hard input; no local variant permitted	ICD s10 change process: a daughter that cannot meet it raises a blocking issue against LUM-CAR-A, it does not implement a variant
P0	Common footprint adopted from ICD s7.1: 100.0 x 80.0 mm, R3.0 corners, 5x M3 (4 corners at 5 mm inset + H5 at (46,74)), 1.6 mm FR4, with a mandatory 30x26 mm RJ45 notch at (6,0)-(36,26)	H1-Q4 closed the stack at 11.0 mm; the carrier's board-edge RJ45 is ~15 mm tall so the notch is a hard mechanical stop and the anti-180-degree interlock (ICD s7.4)
P2	Re-baselined onto ICD rev A2 (copied from boards/lumina-carrier/architecture/connector-icd.md)	Two changes bind this board: (1) board-wide HV clearance raised 0.60 -> 0.635 mm per TPS2378 vendor layout guidance, daughters inherit it; (2) new s8.4 - the carrier's TPS16630 load switch has a mandatory 10k SHDN pull-down, so a daughter may assume an unprogrammed/crashed/reset/brownout carrier presents 0 V at J3, not 48 V. Rail limits s6.2 and mechanical s7.1/s7.2 are UNCHANGED.
P2	LED stage stays on +12V (baseline); the P1 '48V is 16x better for dimming slew' argument is WITHDRAWN as an artifact of holding L fixed	Verified algebraically by the orchestrator: with L sized for the same ripple fraction k, t_rise+t_fall = 1/(f*k), independent of Vin. At 700 kHz / k=0.30 both rails give 4.762 us (48V only redistributes 3.81/0.95 us between rise and fall). That is 33.8x the 141 ns PAR-REQ-01 budget, so NO rail or inductor choice reaches the requirement by gating the converter - it needs shunt-FET dimming either way. The 48V branch therefore buys nothing on fidelity and costs 0.635 mm creepage, 100 V caps, bleed path and daughter-owned inrush.

Phase	What	Why
P2	Stackup JLC04161H-3313, 4 layers, 100.0 x 80.0 mm, 1.6 mm	Switch-node containment 11 mm above a live 2.4 GHz antenna (H1-Q8 made Wi-Fi a supported control path), continuous GND reference under jitter-sensitive PWM, B.Cu consumed by two reverse-mounted sockets, ˜120 parts in ˜45 cm2. No controlled-impedance net exists on this board.
P2	Dimming topology: shunt FET per channel across the LED string	Closes the PAR-REQ-01 TIMING wall: TPS92515HV direct PWM floors at 200 ns vs a 141 ns requirement, but its shunt mode is spec'd 10,000:1 = 10.2 ns min pulse. Implemented T ˜ 50-100 ns, 1.4-2.8x margin. The RESOLUTION wall (17 bits needed, LEDC caps at 14) is NOT closable on this board and needs carrier firmware dithering (CR-5).
P2	Design point 150 mA/die, 4 emitter packages, 2S2P per channel; emitters off-board on aluminium MCPCB	Full white = 0.718 A on +12V, 4 % under the sustained ceiling and 2.8x under OCP, so the board is electrically incapable of exceeding its budget with every PWM stuck at 100 %. Off-board MCPCB because 1.6 mm FR4 misses the junction budget by 2-3x. >3 A conditional flag CLOSED (max 0.30 A/channel).
P2	No fifth (amber) channel on rev A; fit BOTH ID mechanisms (ID_ADC bottom-leg divider + I2C EEPROM)	Amber dilutes an af envelope that full white already fills to 96 % of the rail ceiling. ICD s3.3 makes the divider and the EEPROM coexist rather than alternatives - the divider gives board type, the EEPROM per-unit calibration - so PAR-REQ-17 is met without spending a pin.
P2	Q1 CLOSED: enclosure sealed with LED heat conducted through the wall (not vented)	Owner decision: keeps dust out of a basement/garage install over years, and the strobe run independently recommended the same, so both fixtures share one enclosure design. 3.21 W internal -> 38 C air. The wall-conduction path becomes load-bearing for the thermal case and must be written as an explicit testable ENC-x criterion, not an assumption.
P2	Q2 CLOSED: emitter is RGB 3-in-1 + separate white discrete, superseding D3's integrated 4-in-1	Owner decision, taken knowingly against this run's recommendation. Deciding factor: the thermal case now rests on a published 125 C Tj max and 3.5x stock depth rather than on an ASSUMED 12 K/W Rth from a sole-source vendor publishing neither Rth nor Tj max. Retires the 20 K/W marginality risk this run flagged. Consequences owned here: PAR-REQ-15 white/colour fringing becomes the live risk; the entire sustained-power table is stale and must be re-derived; 145 mA/die is reopened.

Phase	What	Why
P2	Q3 CLOSED: PAR-REQ-01 met by hardware + firmware. Strict reading (5% perceived, gamma 2.2) binds.	Carrier firmware commits to PWM-domain dithering >=3-4 bits (CR-5 accepted). Shunt-FET hardware half stands as designed (T ˜50-100 ns vs 141 ns). FIRMWARE DEPENDENCY must be named in the design doc so it cannot be silently dropped: without the dither the requirement fails by 4.4-8.9x and no hardware on this board can fix it. Dither must be PWM-domain, not 60 fps frame-domain - a 4.4% dither at 60 Hz breaches IEEE 1789 NOEL on its own.
P2	Q4 CLOSED: same-reel sourcing; 24C32 EEPROM fitted but shipped empty; PAR-REQ-16 and PAR-REQ-17 to be formally amended with original text quoted	No measurement instrument exists today, so colour matching rests on same-reel binning. The EEPROM is fitted so that decision stays reversible. Amending rather than quietly failing PAR-REQ-16 keeps the record honest about what was changed by decision.
P2	PAR-REQ-16 and PAR-REQ-17 formally amended (AMD-01, AMD-02 in requirements.md s0), original text quoted	AMD-01: a chromaticity/flux bin is not an orderable attribute on LCSC/JLC and the selected vendor publishes none, so the BOM line was unbuildable; replaced by single-reel procurement for the whole build. This is a REDUCTION in guarantee, not an equivalent - same-reel is an empirical correlation, a published bin is a contractual limit - so PAR-REQ-06 is carried at reduced confidence with bench verification. AMD-02: hardware (divider + 24C32) is fitted; the calibration DATA is deferred because no colorimeter exists for this build.
P2	The RJ45 notch is the coordinator's to implement via a central board_init –cutout flag. This run does NOT hand-edit Edge.Cuts and does NOT shrink the outline.	Both daughters need the notch, so it is being added centrally rather than worked around per-run. The notch stays a hard keepout in constraints.json meanwhile. Coordinator will confirm the flag exists before this run reaches P5. Supersedes the OPEN-7 remediation (hand Edge.Cuts edit) recorded at H1; the OPEN-7 coordinate-translation trap under fixed-outline board_init still stands and must be re-raised at H2.
P3	P5 outline procedure fixed: board_init –outline 100x80 –corner-radius 3 –mounting-holes 4 –cutout 6,0,30,26, then verify geometry against an expected area of 7212.274 mm2	Coordinator added –cutout centrally (commit 18613d3) so both daughters share it. Expected area derived here so P5 has a number to check rather than eyeballing: 100x80 = 8000, minus the 30x26 notch = 780, minus 4 corners at r=3 = 4 x r^2(1-pi/4) = 7.7257. The outline_origin translation is the trap that would otherwise silently misplace every keepout.
P3	Emitter set: 4x XL-HD6070RGBC-A46L-BD (C22434861) + 4x XL-HD6070UWC-A4-BD (C48586656); 2S2P, 150 mA/die, 300 mA/channel. Headline 0.717 A = 95.6% of the 0.75 A ceiling.	Both live-verified via parts_search this session and spot-checked independently by the orchestrator. The headline survives the package change because BOTH packages publish identical Vf ranges (R 1.8-2.4, G/B/W 2.8-3.4), so the 2S2P/150 mA arrangement falls out the same way - a derived result, not a copy-forward. All nine power rows independently recomputed by the orchestrator and reproduce exactly.

Phase	What	Why
P3	150 mA/die confirmed over 145 (H1-P7 re-derived, not carried forward)	Every genuine HARDWARE ceiling passes at both: 2.8x on the 2.0 A OCP, 3.4x on the TPS2378 0.85 A MINIMUM limit. The only figure 150 fails is the ICD 8.5 W total, which ICD s6.2 itself calls a firmware-governed average, not a trip. Flips to 145 if the carrier owner declares 8.5 W a hardware limit.
P3	Sense resistor 0.75 ohm (not 0.68), nominal 275 mA/channel - the COMPLIANT value is the default	P3 found that 0.68 ohm with IADJ tied to VCC gives a +-5.6% threshold band -> ILED 284-324 mA -> 0.775 A worst case = 103% of the ICD 0.75 A +12V sustained ceiling. Orchestrator reproduced it and also costed the proposed IADJ divider: the divider tightens the band but lowers the threshold too, landing at essentially the same light (0.928 vs 0.917 of nominal) for 8 extra resistors - REJECTED. Compliant-by-construction is the right default because the stuck-PWM backstop is exactly the case where firmware is dead and cannot govern the average. Costs 8.3% light vs the H1 point. Reversible: BOM value only, zero layout impact, changeable to P9. Flagged for H2.
P3	Shunt FET switched to TOSHIBA SSM3K2615R,LXGF (C22371361, SOT-23F)	Closes two P3 open items at once. Ciss 150 pF into the 150 R gate resistor is tau ˜22 ns, so gate edges land in the tens of ns against PAR-REQ-01's 141 ns pulse - the previously selected part was estimated at 100-140 ns, which would have eaten the entire timing budget. And Rds(on) 580 mohm is specified AT 3.3 V, so it is guaranteed on at our gate drive rather than inferred. Also resolves the no-first-tier-brand-in-stock risk. Dissipation 69 mW at 0.345 A peak. Package is SOT-23F not SOT-23 - fp_verify needs extra attention.
P3	Keep the THT reverse-mounted ICD sockets; REJECT the HC-PM254 SMD equivalents	ICD s5.2 derives its 0.84 mm pad-to-pad creepage from the THT land pattern (1.70 mm annulus on a 1.02 mm drill), and that geometry is the carrier owner's frozen number. An SMD land pattern changes it and would require the ICD to be re-issued. Two hand-solder operations across 8 boards is the cheaper side of that trade.
P3	Re-synced ICD copy again; daughters MUST use a 1.70 mm annulus on the connector pads (1.10 mm drill, 0.300 mm/side ring)	The carrier's P3 library work corrected the drill from the assumed 1.02 mm to an actual 1.10 mm, and normalised J3's annulus from 1.80 to 1.70 mm so both frozen connectors share one land pattern. The 0.84 mm pad-to-pad gap and the 1.32x clearance conclusion are unchanged, but the annulus is now an explicit daughter requirement - this is a librarian/fp_verify checkpoint, not just a note.
P3	0.75 ohm sense resistor ENDORSED by the coordinator; C2934269 recorded as the explicit 0.68 ohm alternate	Coordinator: 'a margin that only holds while software works is not a margin.' Reversion to 0.68 ohm stays live as a one-code swap through P9.

Phase	What	Why
P3	CR-1 CLOSED: R206 = 4.7 kOhm, 1% REQUIRED (not preferred). LUM-PAR-A is ID_ADC code 2, V_ID = 1.055 V.	ICD s3.4 rev A3, NORMATIVE. Verified: 3.3 x 4700/(4700+10000) = 1.0551 V. 1% is required because adjacent-code separation is 0.353 V against a +-0.05 V divider error from 1% parts plus a +-0.15 V firmware window - a 5% part eats the margin. Cited by section+amendment revision, not the header (the header reads A3 while its own parenthetical lists amendments through A6). The strobe run found its placeholder was set to THIS board's code and would have shipped announcing itself as the par - the value is functional, not cosmetic.
P3	CR-6 CLOSED by ICD s7.7 rev A5 (NORMATIVE): sealed + wall-conducted, internal air 32-33 C at a 25 C room	The carrier re-derived and reconciled. With the LED wall path WORKING only 1.894 W enters box air (6.8-8.1 K rise); with it FAILING 8.5 W enters and gives 56-62 C - which reproduces the rev A2 af figure to the degree. So the af number was never miscalculated, it was the LED-heat-stays-in-box case at a 25 C room with NEITHER assumption written down. The at figure WAS a genuine error, exactly as this run found. This run's 89-115 C Hoffman/Rittal figure is now recorded as the CONSERVATIVE BOUND and the ICD explicitly says do not average it. Budget: 10.5-12.5 W of box air at a 25 C room holding air to 70 C.
P3	PWM allocation NORMATIVE (ICD s3.5 rev A3): LUM-PAR-A runs PWM0-3 on ONE timer, 13-bit / 9.766 kHz, with a 90 degree phase stagger. CR-3 (14-bit/4.883 kHz) is GRANTED and available opt-in; CR-4 (stagger) is GRANTED.	The prior ICD claim that PWM0-3 sit on timer 0 and PWM4-7 on timer 1 is WITHDRAWN - the carrier does not hard-assign timers; each daughter declares its own map and firmware applies it from the ID_ADC code. Three timers remain free for this board. The 14-bit option doubles the driver timing budget from 141 ns to 281 ns and makes firmware dithering physically reproducible - it costs camera-flicker margin and must be re-tested against a phone camera on the first prototype.
P3	ADOPT the 14-bit / 4.883 kHz PWM mode (CR-3, granted) as this board declared timer map, instead of 13-bit / 9.766 kHz	It doubles the driver timing budget from 141 ns to 281 ns, which converts the shunt-FET timing risk from unverified-and-probably-violated into comfortable margin against the only switching number that exists (toff 150 ns typ). It ALSO doubles the LSB duration, which is what makes firmware PWM-domain dithering physically reproducible - spec-dimming s2.7 ranks 1 and 2 as a package, not alternatives. Cost is camera-flicker margin: band contrast rises from 0.61% to 1.23% at 60 fps (still under the visibility threshold) and 2.46% to 4.92% at 240 fps - and H1-P9 already recommends declaring 240 fps out of scope. ICD s3.5 makes the mode opt-in per daughter and requires a phone-camera re-test on the first prototype.

Phase	What	Why
P3	APPROVED library edits: J3 annulus 1.80->1.70; J4 annulus 1.60->1.70 and drill 1.050->1.100; inductor footprint swapped to KiCad stock Inductor_SMD:L_Cenker_CKCS4030; SMF15A accepted as-pulled with a note	J3/J4: ICD-01 s5.2 mandates 1.70 mm annulus on a 1.10 mm drill and daughters must conform - the carrier already normalised its mating DS1021 pair the same way (its lib/EDITS.md edit 1). Note the as-pulled J3 gap of 0.74 mm still CLEARS the 0.635 mm creepage requirement at 1.165x, so this is an ICD-conformance fix, not a safety fix - but conformance is the point of a frozen interface. Inductor: as-pulled land is 3.20 mm centres against a 2.60 mm datasheet land, a 0.60 mm error with pads hanging 0.40 mm past the body; KiCad stock at 2.70 is a strict superset of the datasheet land. SMF15A: 3.26 vs 3.00 pitch is a 0.13 mm outboard shift with pad size EXACTLY matching the datasheet - solderable, accept.
P4	P4 schematic-block agents run at opus/high instead of the tier table's fable/medium	Fable 5 credits exhausted account-wide. The skill's own tier table ranks fable/medium ABOVE opus/high, so opus is the nearest substitution that is not the silent downgrade the spawn rules forbid; user chose it over a cheaper mixed or all-sonnet split with the drivers/thermal safety content in view
P4	The power sheet owns ALL FOUR PWR_FLAGs (GND, +12V, +3V3, +48V_SW); every other sheet places consuming power symbols only	each rail enters the board through J3's passive pins on the power sheet, so that is the one honest driver; PWR_FLAG's pin is power_out and a second flag on the same rail collides power_out<->power_out and fails the erc gate
P4	J5 entry-direction question CLOSED: S10B-PH-SM4-TB is SIDE ENTRY and is the CORRECT part - no swap-part	datasheet p3 splits the model table into Top entry (B2B..B16B-PH-SM4-TB) and Side entry (S2B..S15B-PH-SM4-TB) columns, repeated twice more for the THT and LIF variants: the entry code is the LEADING letter, and the B in '10B' appears in every model of the series. Side entry is also right for this board - the module mounts to the enclosure wall so the harness leaves laterally regardless, and J5 is on B.Cu inside the ICD's 11.0 mm mezzanine gap where a top-entry PHR-10 housing plus bend radius would foul the carrier

Phase	What	Why
P4	The /PWM0..3 nets MUST carry the canonical root-level names /PWM0../PWM3; preferred mechanism is exposing them as hierarchical pins from control, with re-spelling constraints.json to /control/PWMn as the sanctioned fallback if that is not ERC-clean	sheets.md s2 states these are the names the schematic must produce, and constraints.json.high_speed feeds SEVEN consumers (check_return_path, planes_gen, route_critical, rules_gen, stitch_vias, constraints_lint, netlist_audit) - a mismatch silently drops the jitter-sensitive PWM nets from every one of them, on the board whose hardest requirement is the PAR-REQ-01 141 ns timing wall. It also fails netlist_audit outright (missing_net is an error), so it must be settled before the root build. sheets.md is internally inconsistent here - it lists PWM in the INTERFACE net table while its own From->to column calls them control-internal - so the tie-break is which artifact ends up TRUE: name-match is required either way, and the hier-pin route keeps one uniform convention across all root nets
P4	REVERSAL of the earlier P4 ruling: /PWM0..3 keep their real netlist names /control/PWM0..3, and constraints.json high_speed was re-spelled to match. The preferred hier-pin mechanism was tried, measured and REJECTED	the hier-pin route works electrically (netlist carried /PWM0 = J4-1 + R202-1 + U202-1) but is NOT ERC-clean: 4x label_dangling, because the root subgraph for those nets is wire + label + exactly ONE sheet pin and KiCad 10.0.3 requires a real connection point. Two probes settled the rule: a second same-named root label does NOT satisfy it (8 errors, both flagged), while one TestPoint pin per root PWM wire takes the identical project to 0/0 - so buying the bare name would mean adding four PARTS to the root, a design change rather than a stitch fix. Every other root net escapes this only because two children expose it. lumina-carrier never exercised it: all 70 of its root labels appear 2-3 times. Outcome is equivalent for the seven high_speed consumers, which match on exact name and do not validate name shape
P4	H2: emitter-hot hysteresis band ACCEPTED at 8.6 K (trip 89.9 C, release 81.2 C) against a 15 K aspiration; R207 stays 100k	trip is exact on spec at 89.9 C and only the release sits high. Widening needs a smaller feedback resistor, which trades directly against the emitter-OPEN channel margin - and emitter-open is the fail-safe that catches the broken harness this block exists for. Board-hot is already on spec at 110.2/92.9. The R207 route was recommended against and rejected: the thermal rework removed its justification (released FAULT now >=2.81 V in every corner incl. unmated-hot, which used to sag to 1.80 V below U201 VIH) and a stronger pull-up would shift every trip point, forcing a full MNA re-solve for no remaining benefit

Phase	What	Why
P4	Optical budget to be RE-DERIVED before P5, on the OPEN-8 refutation	both emitters are 140 deg 2Theta1/2 per their own datasheets; the LCSC parametric record for the white part contradicts its own datasheet. The diffuser strength - and therefore its 30-45 pct flux cost, and the compounded 36-50 pct reduction against the H1-approved flux - was sized on a 140/120 deg delta that does not exist. Owner chose to re-derive now rather than defer, since that number is what the fixture gets judged on and the correction may recover real light
P5	Stackup changed JLC04161H-3313 -> JLC04161H-1080B	board_init REFUSED -3313 by name: stackups.yaml marks it available:false - it is a PHANTOM that JLC has never offered for 4L/1.6mm/1oz/0.5oz, and it was stackups.yaml own 4-layer default until it was retired 2026-08-06, which is exactly why the P2 architecture (2026-07-28) picked it. 1080B is JLCs only real lamination in this class, live-verified 2026-08-06. FORCED, not a judgment call - the original is unorderable. Low impact here: this board has NO controlled-impedance net, so the consequence that hit lumina-carrier (100 ohm MDI solved on -3313 measuring 105.5 ohm on real 1080B geometry) does not apply. Stack total 1.5908 -> 1.6542 mm
P5	.kicad_dru HV clearance hand-edited 0.6000 -> 0.6350 mm on +48V_SW	ICD-01 s5 TRAP-1 and stackup.md s7 rule 1 both state it explicitly: the BINDING requirement is 0.635 mm (TPS2378 layout section), while rules_gen emits 0.6000 mm - the IPC-2221B B2 51-100 V FLOOR. check_creepage.py only ever demands 0.60, so a 0.635 layout passes it BY CONSTRUCTION and the extra 0.035 mm is enforced ONLY by this DRU rule. stackup.md: do not skip it because P8 goes green. RE-APPLY after any rules_gen re-run - rules_gen overwrites the file
P6	ICD s7.2 RE-ISSUED from the carrier as-placed geometry: J3 anchor board-local (23.000, 76.000) pad1 (15.380, 77.270); J4 anchor (71.000, 76.000) pad1 (57.030, 77.270); H5 (46.000, 74.000) unchanged. The daughter places to THESE, not to the old nominal table.	ICD s7.2 was the one section never frozen at H1 and explicitly required the carrier owner to compare placed positions and re-issue before any daughter run; lumina-carrier reached P10 without that step. Measured delta is +7.970 mm in Y on BOTH connectors identically (X agrees within 0.08/0.27 mm, and H5 matches exactly, confirming a shared origin convention) - the carrier annealer moved the connector row and the table was never corrected. Owner ruled the as-built carrier is ground truth: it is at P10 and fab-ready, so moving it is the expensive direction. Placing this daughter on the stale table would have put its sockets ˜8 mm from the carrier headers - a non-mating pair, and invisible to every gate because nothing in the pipeline compares two boards.

Phase	What	Why
P6	J3 and J4 net-to-pin maps SWAPPED row A <-> row B on this board (ICD s3.1/s3.2 transcribed MIRRORED, not literally)	the daughter reverse-mounts both sockets on B.Cu facing down (ICD s7.3), and flipping a 2-row connector to the back swaps its ROWS while columns stay put - measured on the real boards, all 14 J3 pad positions coincide exactly but carrier pin 2k+1 contacts daughter pin 2k+2. ICD s3.1/s3.2 give ONE table with no mirroring instruction and LUM-CAR-A transcribed it literally on its top-side male header, so a literal transcription here crossed 12 of 14 J3 contacts INCLUDING SIX THAT SHORT +48V_SW TO GND, plus 12 live J4 crossings (PWM0<->PWM1, ENABLE<->FAULT, I2C_SDA<->ADC0, and the carriers driven DSPI_SCK onto GND). Re-verified after the swap: 0 crossings on both connectors. Found only because the mated view was checked against the carriers own netlist - erc 0/0, netlist_audit exit 0 and every other gate passed throughout, since every check in ai-ee compares a board against ITSELF
P8	H1 (top-left M3 mounting hole) DELETED from the board; assembly ships on 4 standoffs. ICD-01 re-issued to rev A8.	ICD s7.1 (5 mm corner inset) and s7.6 (RJ45 relief starting at x=6) are mutually incompatible: they leave a 6 mm tongue, and an M3 clearance hole of 1.6 mm radius centred at (5,5) reaches x=6.6, breaching the notch wall by 0.601 mm. Orchestrator-verified: 1.046 of 8.042 mm2 (13.0 pct) off-board, opening a 2.500 mm slot - that corner never had a usable standoff on any daughter. Invisible to every gate because edge_clearance is a COPPER rule and a mounting hole is NPTH. Owner ruled drop it: the lost corner is mostly notch, H2/H3/H4/H5 still bracket the board including the connector edge, and the CARRIER needs no change (its hole simply goes unused), so a board already at P10 is not reworked. Removed via the update_swig worker directly - board_update PLANNER deliberately ignores board_only footprints so it will never plan this, but the worker it drives supports remove_refs. drc_routed and verify both still PASS after removal.

Phase	What	Why
P8	ACCEPT U341 at 7 in-pad thermal vias (8 near-part) against a declared min_vias 9, and D361 at 3 against a declared 4. No further routing cycle.	check_thermal measures every driver at a 7.67 C rise against a 30 C allowance (theta_ja 51.1, 0.15 W) and every catch diode at 7.15 C - roughly 4x margin, so the via-count delta is thermally irrelevant at these dissipations. The blockers are real and both would cost a routing change: U341 8th seat fails min_hole_clearance by 12.6 um because /control/PWM1 runs on B.Cu directly under its exposed pad (a 0.01 mm scan of the whole EP found ZERO legal seats), and D361 is blocked by /EN_OK on In2 crossing under its land. PWM1 is one of the two jitter-critical nets just relocated to fix a return-path defect; moving it again to chase a thermally-unnecessary via is a bad trade. The declared 9 stays in constraints.json as design intent, NOT lowered to match the board - lowering a spec to fit the artifact is how a requirement quietly disappears.

Sheet plan

lumina-par (LUM-PAR-A) - hierarchical sheet plan

Five sheets under a root that contains nothing but the sheet symbols and the inter-sheet wiring. Interface nets become hierarchical pins at P4 and placement groups at P6.

Net-naming contract (binding on P4). KiCad names a net from a hierarchical label alone as `/<sheet>/<LABEL>`. To produce the canonical names below, **P4 must place a root-sheet local label on every inter-sheet wire**, spelled exactly as in s2. `constraints.json` uses these names and P5-P8 will silently no-op on any net whose name does not match. Power nets are bare global symbols (`+3V3`, `+12V`, `+48V_SW`, `GND`); every other inter-sheet net carries a leading `/`.

—

1. Sheets, refdes ranges and contents

Refdes are unique across sheets by construction: each sheet owns a hundreds block for every prefix, and each sheet has its own `#PWR` base.

```
| Sheet | Blocks | Refdes block | ‘#PWR‘ base |
|---|---|---|---|
| ‘power‘ | B1 - rail entry, bulk, branch-B front end (DNP), test points, H5 |
**100-199** | **100** |
| ‘control‘ | B2 - ENABLE gating and fail-safe logic; B5 - ID and calibration |
**200-299** | **200** |
| ‘drivers‘ | B3 - the four constant-current channels and their shunt FETs |
**300-399** | **300** |
| ‘thermal‘ | B4 - NTC dividers, window comparator, FAULT driver | **400-499** |
**400** |
| ‘led_if‘ | B6 - LED harness header, TVS clamps, thermocouple pads | **500-599**
| **500** |
```

Exception: J3, J4 and H5 keep the refdes the ICD assigns them (ICD s7.2 / s7.5 name them J3, J4 and H5, and the carrier's placement tables refer to them by those names). They are exempt from the hundreds blocks. J5 is this board's own harness header and follows the same connector sequence.

1.1 power (100-199, #PWR 100)

Ref	Part	Note
J3	CONNFLY DS1023-2*7SF11, 2x7 socket	**reverse-mounted, bottom side, facing down** (ICD s7.3). Pin 1 silkscreened with a triangle
H5	MountingHole_3.2mm_M3	(46, 74), ICD s7.5. Not a ‘board_init‘ hole - added here so it carries a refdes
C101, C102	22 uF / 25 V X7R	‘+12V‘ bulk
C103, C104	10 uF / 16 V X7R	‘+3V3‘ bulk
C105	100 nF	‘+3V3‘ HF
TP101-TP103	test pads	‘+12V‘, ‘+3V3‘, ‘+48V_SW‘. **TP103 carries the ICD s9 bench-hazard silkscreen**
Q101, Q102, R101-R104, C106-C108	**branch-B front end - DNP on branch A**	100 V P-FET hot-swap, Cgd dV/dt network, gate pull-down N-FET from ‘/EN_OK‘, 100 k **0805** bleed, 2x 10 uF/100 V bulk. R104 is 0805 now so branch B needs no footprint change (ICD s5.4)

1.2 control (200-299, #PWR 200)

Ref	Part	Note
J4	CONNFLY DS1023-2*12SF11, 2x12 socket	reverse-mounted, bottom side, facing down
U201	SN74LVC1G08 (SOT-23-5)	‘/EN_OK = ENABLE AND FAULT‘
U202	74LVC00A quad NAND (SO-14)	‘/SHUNTn = NOT(PWMn AND /EN_OK)‘ - one package covers all four channels
U203	24C32-class I2C EEPROM	**address 0x50**, WP to GND. **No I2C pull-ups on this board**
R201	100 k	**ENABLE pull-down at the connector end, before any series element** (ICD s8.2)
R202-R205	4x 100 k	PWM0-3 pull-downs at J4 - an undriven carrier must not float a gate input high
R206	**VALUE TBD**	‘ID_ADC‘ bottom leg. **Allocated by the carrier owner, not chosen here** (ICD s3.3). P4 blocker - ‘decisions.md‘ OPEN-2
R207	100 k	local ‘FAULT‘ pull-up to ‘+3V3‘, so the node is defined when unmated
R208, R209	0 R / 10 k	EEPROM A0-A2 and WP strapping
R214-R217	4x 0 R	‘/EN_OK -> /DRV_ENn‘ links, **fitted by default**
U204, R210-R213, C210-C213, D201-D204	**converter-idle one-shot - DNP option**	74LVC14 hex Schmitt + retriggerable RC + diode per channel. Populate to save ~0.8 W at saturated colours (‘power_tree.md‘ s2); if populated, remove R214-R217
C201-C205	decoupling + ‘ID_ADC‘ 100 nF	

1.3 drivers (300-399, #PWR 300) Four identical channels; channel n is based at 300 + 20n. Channel 0 shown; channels 1-3 are U321/U341/U361 and so on.

Ref	Part	Note
U301	TPS92515HVDGQR class (MSOP-10-EP)	5.5-65 V, 2 A internal FET, shunt-PWM capable. Exposed pad on a >= 9-via array
L301	47 uH shielded	sized for **ripple, not slew** - 90 mA (30 %) at ~700 kHz
D301	**60 V** Schottky, 1 A	branch A. Low Vf matters: it dominates the 0.20 W idle loss. Branch B needs a 100 V part
R301	sense resistor	sets 300 mA (af, branch A). **The at upgrade is this value**
Q301	60 V logic-level N-FET	shunt across the string. **60 V now, not 30 V**, so the branch-B / ‘at‘ 8S string (27.2 V) needs no change

```
| R302 | 150 R | shunt gate series - bounds the 74LVC00 output to 22 mA, ~30 ns
edges |
| R303 | 100 k | shunt gate pull-down - an unpowered gate IC must not leave a
FET half-on |
| C301 | 4.7 uF / 25 V | VIN local (100 V on branch B) |
| C302 | 100 nF | VIN HF, at the pin |
| C303, R304 | **DNP** | SW-node RC snubber footprint |
```

There is no output capacitor across any LED string - a shunt FET dumps it every PWM cycle (`blocks.md` B3 rule 1). Do not let a reviewer add one.

1.4 `thermal` (400-499, `#PWR` 400)

```
| Ref | Part | Note |
|---|---|---|
| U401 | quad open-drain comparator, LMV339 / TLV3704 class | rail-to-rail
input, <= 50 uA/channel.  **Open-drain outputs are mandatory** - they wire-OR
onto ‘FAULT‘, which must never be driven high |
| RT401 | 10 k B3950 NTC, 0603 | **on the copper of the hottest driver stage,
outside the DC-DC hot zone (2,46)-(36,68)** |
| R401, R403 | 2x 10 k | NTC divider top legs (board, module) |
| R402, R404 | <= 1 k | ADC series - keeps total source impedance <= 6 kohm
against the ICD’s 10 kohm |
| R405-R412 | reference ladder + hysteresis | ratiometric off ‘+3V3‘, so
thresholds track the carrier’s ADC reference |
| C401-C403 | decoupling + NTC filter | filter caps sized so R402/R404 x C stays
a slow-signal filter, not a pulse filter |
```

Four comparator channels: **emitter hot (90 C, release 75 C) / emitter open / emitter short / board hot (110 C, release 95 C)**. The window (open + short) converts the most likely fault in an off-board module - a broken harness wire, which reads as *cold* in either divider orientation - from fail-dangerous to fail-safe (D-T17, E-10).

> **P8 SPICE AMENDMENT 2026-08-08 - the fault window has THREE functionally > distinct channels, not four.** An independent SPICE bench (which reproduced > the hand MNA solve to within 0.4 K on every channel) measured **CMP3 > (emitter short) as functionally redundant with CMP1**: its tap sits at > 0.311 V, BELOW CMP1’s 0.427 V, and both detect ”sense node low”, so CMP1 > always asserts `/FAULT` first. CMP3 measured a degenerate **1.1 mV band** and > can never define a `/FAULT` edge. This is a leftover from the inverted- > orientation spec that the blocks.md B4 amendment already corrected. > It is harmless and free - worth keeping as redundancy against a single > comparator failing - but **no document should claim four independent fault > detectors.** Measured trip/release: emitter hot 89.67/81.26 C (+8.41 K), > emitter open -22.30/-3.82 C (+18.47 K), board hot 109.80/92.90 C (+16.90 K).

1.5 `led_if` (500-599, `#PWR` 500)

```
| Ref | Part | Note |
|---|---|---|
| J5 | 10-way latched wire-to-board, 2.0 mm (JST PH class) | 4 anode + 4
per-channel GND return + ‘/NTC_LED‘ + dedicated NTC sense return |
| D501-D504 | TVS, SOD-123 | **15 V standoff on branch A, 33 V on branch B** -
same footprint, a BOM value change |
| TP501, TP502 | bare-copper thermocouple pads | bring-up thermal verification
(L-13).  ICD s9 bench-hazard silkscreen |
```

No 48 V net reaches this sheet, ever (D-T13). That is what keeps the 0.635 mm clearance rule, the 100 V capacitor rule and the 0805-minimum resistor rule off the harness and the module.

—

2. Interface nets - canonical names

These are the names the schematic **must** produce. `constraints.json` and every P5-P8 check key off them.

Net	Type	From -> to	Note
‘+48V_SW‘	power	‘power‘ (J3 1/3/5)	**landed, not tapped** on branch A. Declared in ‘voltages‘ at 57 V so ‘check_creepage‘ enforces the clearance
‘+12V‘	power	‘power‘ -> ‘drivers‘	0.717 A max (‘power_tree.md‘ s3, revision B)
‘+3V3‘	power	‘power‘ -> ‘control‘, ‘thermal‘	<= 5 mA
‘GND‘	power	everywhere	In1 plane
‘/PWM0‘ .. ‘/PWM3‘	in	‘control‘ (J4) internal	3.3 V CMOS. **No RC filter; if any network, tau <= 14 ns**
‘/ENABLE‘	in	‘control‘ (J4) internal	100 k pull-down here
‘/FAULT‘	bidir, open drain	‘thermal‘ -> ‘control‘ -> J4	**never driven high**; wire-OR’d with the carrier eFuse fault
‘/EN_OK‘	internal	‘control‘ -> ‘power‘ (branch-B gate)	‘ENABLE AND FAULT‘
‘/DRV_EN0‘ .. ‘/DRV_EN3‘	out	‘control‘ -> ‘drivers‘	driver PWM/EN pins. Tied to ‘/EN_OK‘ through 0 R by default; driven by the one-shot if populated
‘/SHUNT0‘ .. ‘/SHUNT3‘	out	‘control‘ -> ‘drivers‘	shunt-FET gates. ‘NOT(PWMn AND /EN_OK)‘
‘/LED0_A‘ .. ‘/LED3_A‘	out	‘drivers‘ -> ‘led_if‘	channel anodes, 0.30 A each
‘/NTC_LED‘	in	‘led_if‘ -> ‘thermal‘	module NTC divider node
‘/NTC_BRD‘	internal	‘thermal‘	board NTC divider node
‘/ADC0‘	out	‘thermal‘ -> ‘control‘ -> J4	emitter temperature to the carrier. Source impedance <= 6 kohm
‘/ADC1‘	out	‘thermal‘ -> ‘control‘ -> J4	board temperature to the carrier
‘/ID_ADC‘	out	‘control‘ -> J4	divider bottom leg; value allocated by the carrier owner
‘/I2C_SCL‘, ‘/I2C_SDA‘	bidir	‘control‘ (J4) -> U203	**no daughter-side pull-ups**

Unused ICD signals, landed on J4 and left unconnected on this board: `PWM4..PWM7` (no fifth channel on rev A), `DSPI_SCK/MOSI/MISO/CSn` (a 4-channel PWM design needs no serial device beyond the EEPROM, and `decisions.md` D2 rejects a local PWM generator). Leave them as no-connects with an explicit ~ ERC directive so P4's ERC does not flag them and so a reviewer can see the decision.

I2C address map (this board owns the whole space; the carrier reserves nothing): **0x50 = 24C32 EEPROM; 0x51-0x57 reserved for its own address pins; nothing else on the bus.** The comparator and both NTCs are analogue.

—

3. Placement groups for P6

Declared in `constraints.json` under `placement.groups`. Each buck channel is one tight cluster; the loop that matters is +12V -> C301 -> U301 -> L301 -> D301.

Group	Anchor	Members
‘ch0‘	U301	L301, D301, R301, R302, R303, C301, C302, Q301
‘ch1‘	U321	L321, D321, R321, R322, R323, C321, C322, Q321
‘ch2‘	U341	L341, D341, R341, R342, R343, C341, C342, Q341
‘ch3‘	U361	L361, D361, R361, R362, R363, C361, C362, Q361

```
| ‘ntc_brd‘ | U301 | RT401 - the board NTC must sit **on** the hottest driver
stage (T-3) |
| ‘gate‘ | U202 | R202-R205, C202 - keep the NAND next to J4’s PWM pins to keep
the jitter-sensitive run short |
| ‘analog‘ | U401 | R401-R412, C401-C403 - **separated from every inductor by >=
10 mm** |
```

`placement.fixed`: **H5**, positioned by an explicit `place_edit` op at (46, 74) in board-local coordinates (translated - see `stackup.md` TRAP 2).

`placement.edges`:

```
| Ref | Edge | pos | Why |
|---|---|---|---|
| J3 | bottom | 0.24 | ICD s7.2 nominal centre x = 24 mm.  **Hint only** |
| J4 | bottom | 0.72 | ICD s7.2 nominal centre x = 72 mm.  **Hint only** |
| J5 | **top** | 0.56 | x ~ 56 mm - the only clear span of the top edge (the
notch occupies x 6-36, the recovery-header keepout x 76-98), and it sits
directly above the driver band so the LED anode runs stay short |
```

J3 and J4’s entries are hints to the annealer, not final positions. Both must be placed with explicit `place_edit` ops at the re-issued ICD s7.2 coordinates and then **locked**, before the annealer runs. Both are **bottom-side, facing down** - set the side at P4 in the footprint or with a `place_edit flip` op, and check pin 1 in the mated view, not from the footprint (pin numbering mirrors across the row on a bottom-side part).

—

4. Sheet-level P4 notes

1. **The DNP option sets must be in the netlist**, not added later: the branch-B front end (`power` Q101/Q102/R101-R104/C106-C108) and the converter-idle one-shot (`control` U204/R210-R213/C210-C213/D201-D204). A DNP part still has pads, so its clearance and area are accounted for at P6/P7 and the option stays a populate change rather than a respin. Mark them `DNP` in the BOM variant field so `bom_cpl` at P9 excludes them. 2. **‘FAULT‘ is open drain.** ERC will want a driving pin somewhere; the comparator outputs plus R207 are it. Do not let a fix loop add a push-pull driver. 3. **No I2C pull-ups.** Fitting them is an ICD violation, and P4’s reflex is to add them. Record it on the schematic. 4. ***‘PWM4..7‘ and ‘DSPI_‘* are deliberate no-connects.** 5. **R206’s value is unknown until the carrier owner allocates it.** Place the part with a `TBD` value field and a schematic note; do not guess a value - that is exactly the silent divergence ICD-01’s preamble forbids.

—

5. P4 AMENDMENTS (2026-08-07)

This file was written at P2. The items below were settled during P4 schematic capture and its adversarial review, and **they supersede the sections above**. `architecture/p4-wiring-notes.md` remains binding for wiring rules.

5.1 s1.2 `control` - new parts

```
| Ref | Part | Note |
|---|---|---|
| **R218-R221** | 4x **10 k** | ‘/DRV_EN0..3‘ fail-safe pull-downs, one per
channel, on the ‘/DRV_ENn‘ side of the R214-R217 links.  **10 k is load-bearing
- do NOT "harmonise" these to the board’s usual 100 k.** The TPS92515HV enable
pin sources up to 25 uA, so 100 k sits at 2.5 V, above the 1.0 V threshold.
Without them, every power-up window with ‘+12V‘ up and ‘+3V3‘ not yet (and any
12 V-only bench bring-up) leaves all four drivers undriven and latching ON - a
```

full-current LED flash, defeating ICD s8.2. Placed with their DRIVER, not with the links. |

R206 is **4.7 k 1 %** (CR-1), not "VALUE TBD". parts.json is re-issued.

5.2 s1.3 drivers - new parts and one changed value The COFF network (C304/R305/R306 per channel, s1.3 had none of it) is joined by the **BOOT network, which s1.3 also omitted and without which the converter cannot switch at all**:

Ref (ch0; +20/channel)	Part	Note
C305	100 nF	BOOT capacitor at pin 4 - the high-side FET's gate supply
D302	1N4148W	BOOT diode, anode on the device's own VCC (pin 2), cathode on BOOT
C306	4.7 uF	VCC decoupler at pin 2, datasheet sec 8.3.6

ROFF2 (R306 etc) is 30 k, NOT 47 k. The original 47 k was sized from TI Equation 9, which assumes VCC charges COFF through ROFF2 alone - false at ROFF1 = 10 k. The shunted COFF node then asymptotes **below** the 1.00 V VOFT at every corner, the off-timer never trips, and shunt-dimming linearity is lost - the exact Figure-43 failure the network was added to prevent. 30 k is solved jointly (dIL_shunt == dIL_normal); 0 of 243 corners now fail. **Do not re-derive this from TI's text alone**: Eq 1 and Eq 8 disagree about the freewheel diode drop and lead to an unsafe ˜38 k.

5.3 s1.4 thermal - range extended, values changed

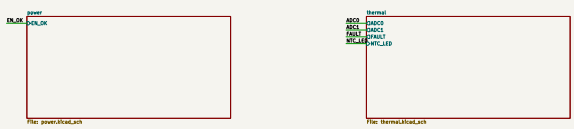
- Ladder + hysteresis is **R405-R416**, not R405-R412.
- **R402/R404 are 150 R, not "<= 1 k"**, and C402/C403 are **4.7 uF 0805, not 100 nF 0603** (footprint change) so the filter stays slow at the lower R.
- **R413/R415** (10 k) series-protect the comparator inputs and **D401** (1N4148W) clamps the PROTECTED comparator node /thermal/CMP_LED to +3V3 - NOT /NTC_LED itself. (Corrected at P8 from the netlist, which is authoritative: R413 sits between them and limits the clamp current to 2.03 mA.) The LM339LV inputs are diode-clamped to GND **only**, with no clamp to V+, so a series resistor alone does not protect them from a harness fault that puts an LED anode (6.8 V nominal, 16.7-24.4 V at the TVS clamp) onto the sense line.
- Hysteresis is returned to the **protected comparator nodes**, not the sensor dividers. See the blocks.md B4 amendment for the solved thresholds.

5.4 s2 net table - /PWM0..3 correction The s2 table's '/PWM0'..'/PWM3' spelling is not what the netlist produces. Those nets are control-internal and come out **'/control/PWM0..3'**; constraints.json.high_speed was re-spelled to match, so all seven high_speed consumers bind correctly. Promoting them to root-crossing names was tried and measured: it is electrically correct but raises 4x label_dangling, because such a net reaches the root as wire + label + exactly ONE sheet pin, and buying the bare name would mean adding four PARTS to the root. /ENABLE, /ID_ADC, /I2C_SCL and /I2C_SDA are in the identical position and are also /control/NAME. No constraint references those four.

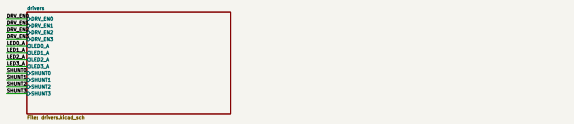
Full architecture narrative (not inlined; see the artifact index): architecture/blocks.md, architecture/decisions.md, architecture/power_tree.md, architecture/stackup.md.

5 Schematic

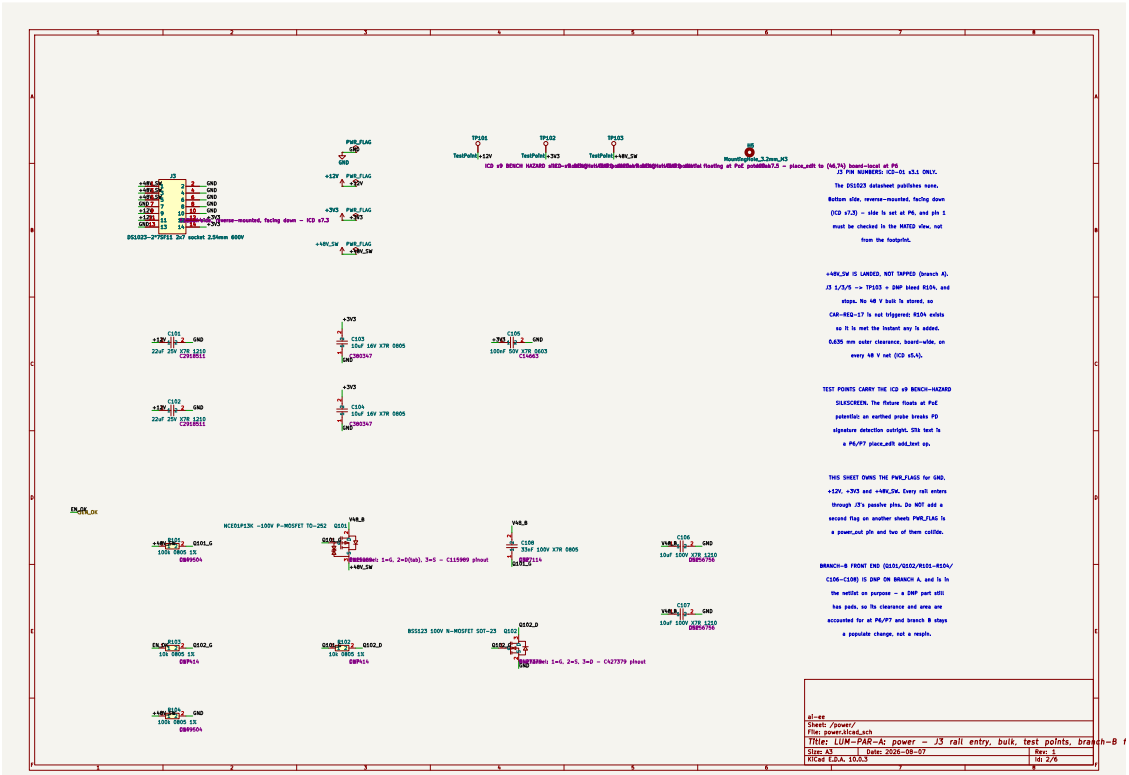
Gate erc (phase P4): **pass** after 2 attempt(s); last 2026-08-07T13:18:16, failing 0 of 0. The full schematic PDF follows.

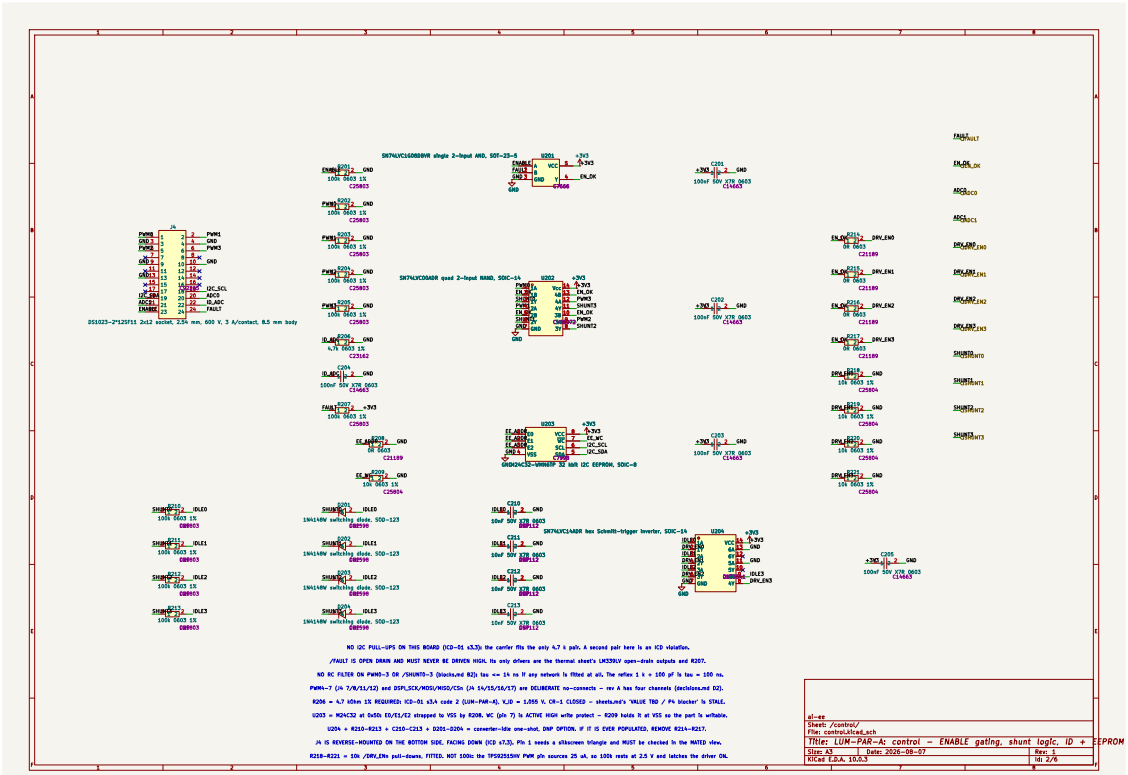


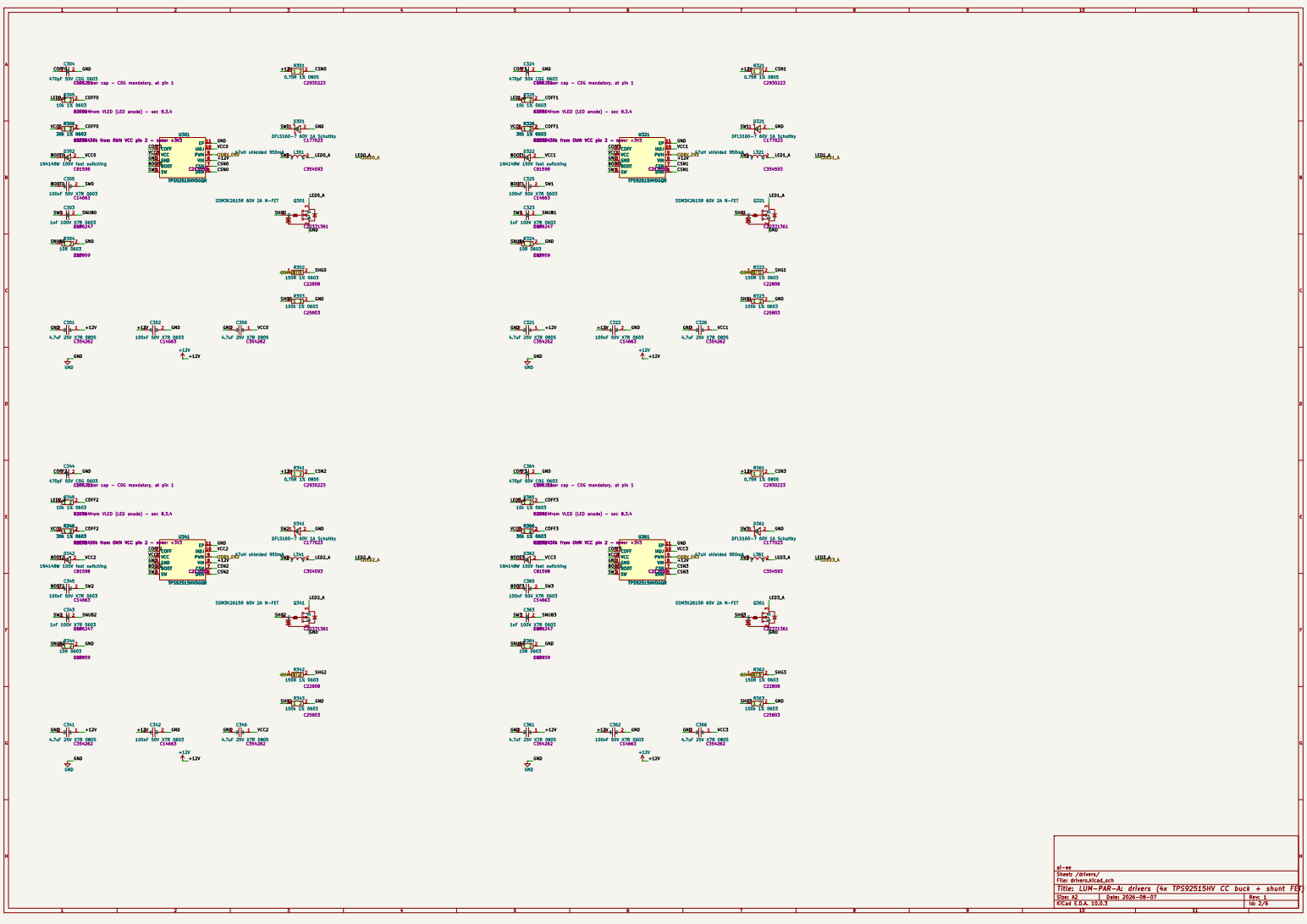




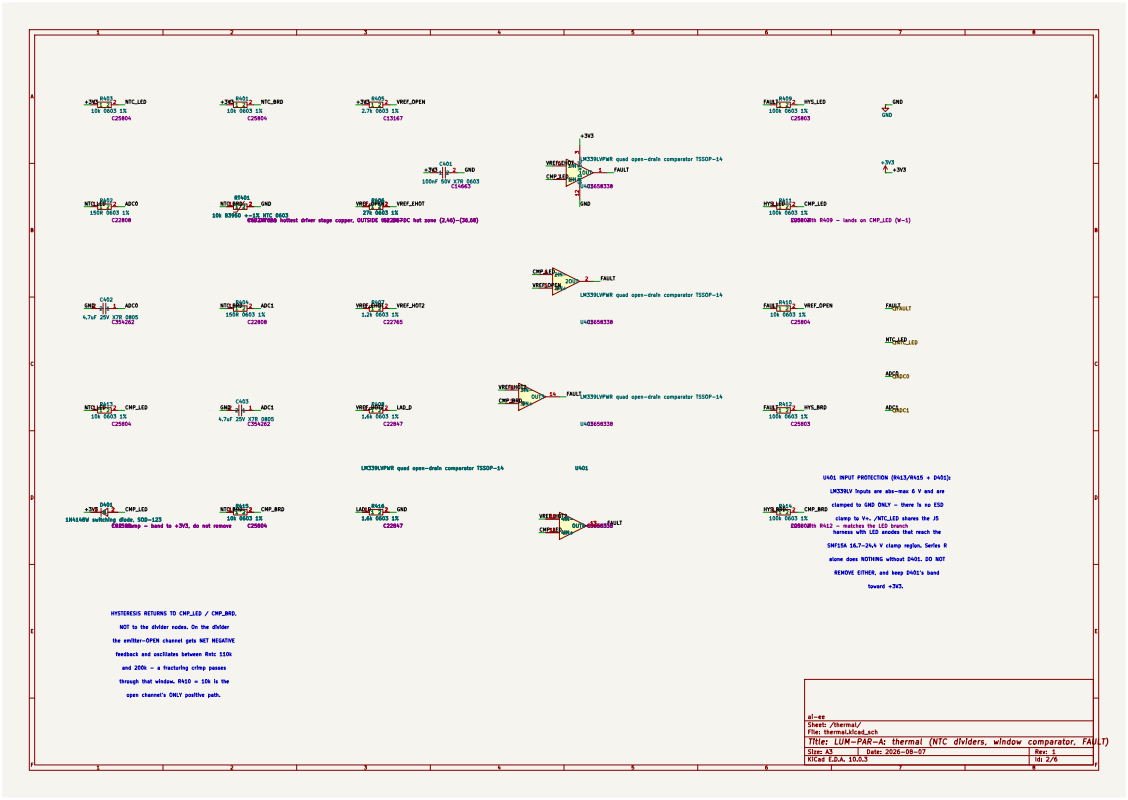
ef-ee	
Sheet: /	
File: lumina-par.kicad_sch	
Title: LUMINA par LUM-PAR-A: root	
Size: A2	Date: 2026-08-07
KiCad E.D.A. 10.0.3	Rev: 1 Id: 1/6

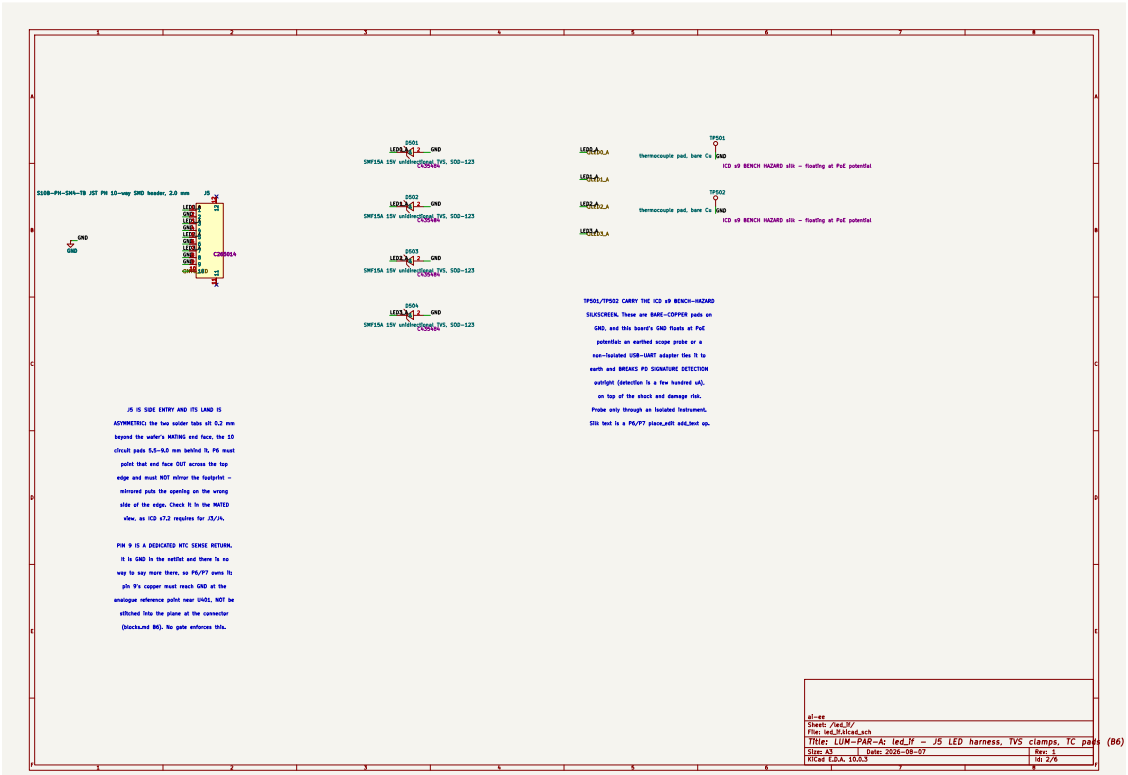

TP101
TestPoint +12V
TP102
TestPoint +3V3
TP103
TestPoint +48V_SW
H5
MountingHole_3.2mm_M3
ICD v9 BENCH HAZARD
Mounting hole floating at PoE potential — 7.5 — place_edit to (46.74) board-local at P6
J5 PIN NUMBERS: ICD-01 v3.1 ONLY.
The DS1023 datasheet publishes none.
Bottom side, reverse-mounted, facing down
(ICD v7.3) — side is set at P6, and pin 1
must be checked in the MATED view, not
from the footprint.
J3
DS1023-2*7SF11 2x7 socket 2.54mm 600V
PWR_FLAG
GND
+12V
+3V3
+48V_SW
+48V_SW IS LANDED, NOT TAPPED (branch A).
J3 1/3/5 -> TP103 + DNP bleed R104, and
stops. No 48 V bulk is stored, so
CAR-REQ-17 is not triggered; R104 exists
so it is met the instant any is added.
0.635 mm outer clearance, board-wide, on
every 48 V net (ICD v5.4).
C101
22uF 25V X7R 1210
C2918511
C103
10uF 16V X7R 0805
C19637
C105
100nF 50V X7R 0603
C14663
C102
22uF 25V X7R 1210
C2918511
C104
10uF 16V X7R 0805
C19637
TEST POINTS CARRY THE ICD v9 BENCH-HAZARD
SILKSCREEN. The fixture floats at PoE
potential: an earthed probe breaks PD
signature detection outright. Silk text is
a P6/P7 place_edit add_text op.
THIS SHEET OWNS THE PWR_FLAGs for GND,
+12V, +3V3 and +48V_SW. Every rail enters
through J3's passive pins. Do NOT add a
second flag on another sheet; PWR_FLAG is
a power_out pin and two of them collide.
EN_OK
NCEO1P13K -100V P-MOSFET TO-252
Q101
Q101_G
Q101_D
Q101_S
R101
10R 0805 1%
C0805R0A
R102
10R 0805 1%
C0805R0A
R103
10R 0805 1%
C0805R0A
VAB_B
C109
33nF 100V X7R 0805
C0805F114
C106
10uF 100V X7R 1210
C0805F118
C107
10uF 100V X7R 1210
C0805F118
BRANCH-B FRONT END (Q101/Q102/R101–R104/
C106–C108) IS DNP ON BRANCH A, and is in
the netlist on purpose — a DNP part still
has pads, so its clearance and area are
accounted for at P6/P7 and branch B stays
a populate change, not a respin.
BSS123 100V N-MOSFET SOT-23
Q102
Q102_G
Q102_D
Q102_S
R104
10R 0805 1%
C0805F114
al-ee
Sheet: /power/
File: power.kicad_sch
Title: LUM-PAR-A: power — J3 rail entry, bulk, test points, branch-B f
Size: A3
Date: 2026-08-07
Rev: 1
KiCad 8.0.4, 10.0.3
Id: 2/6


SN74LVC1G08DBVR single 2-input AND, SOT-23-5
SN74LVC08ADR quad 2-input NAND, SOIC-14
SN74LVC1A04DR hex Schmitt-trigger inverter, SOIC-14
M24C32-DRMN6TP 32 kbit I2C EEPROM, SOIC-8
DS1023-2*12SF11 2x12 socket, 2.54 mm, 600 V, 3 A/contact, 8.5 mm body
NO I2C PULL-UPS ON THIS BOARD (ICD-01 s3.3): the carrier fits the only 4.7 k pair. A second pair here is an ICD violation.
/FAULT IS OPEN-DRAIN AND MUST NEVER BE DRIVEN HIGH. Its only others are the thermal sheet's LMZ35LV open-drain outputs and R207.
NO RC FILTER ON PWM0–3 OR /SHUNT0–3 (blocks,md 82): tau <= 1% ns If any network is fitted at all. The reflex 1 k + 100 pF is tau = 100 ns.
PWM4–7 (J4 7/8/11/12) and DSPI_SCK/MOSI/MISO/CS0 (J4 14/15/16/17) are DELIBERATE no-connects — rev A has four channels (decisions.md D2).
R206 = 4.7 kOhm 1% REQUIRED: ICD-01 s3.4 code 2 (LUN-PAR-A). V_ID = 1.666 V. CR-1 CLOSED — sheets.md's 'VALUE TBD / P4 blocker' is STALE.
U203 = M24C32 at 0x50. E0/E1/E2 strapped to VSS by R208. WC (pin 7) is ACTIVE HIGH write protect — R209 holds it at VSS so the part is writable.
U204 = R210–R213 + C210–C213 + D201–D204 = converter-ide one-shot. DNP OPTION. IF IT IS EVER POPULATED, REMOVE R214–R217.
J4 IS REVERSE-MOUNTED ON THE BOTTOM SIDE, FACING DOWN (ICD s7.3). Pin 1 needs a silkscreen triangle and MUST be checked in the MATED view.
R218–R221 = 10k /DRV_ENx pull-downs, FITTED. NOT 100k: the TPS92551HV PWM pin sources 25 uA, so 100k rests at 2.5 V and latches the driver ON.
al-ee
Sheet: /control/
File: control.kicad_sch
Title: LUN-PAR-A: control — ENABLE gating, shunt logic, ID + EEPROM
Size: A3
Date: 2026-08-07
Rev: 1
KiCad 8.0.4, 10.0.3
Id: 2/6



Sheet /drivers/	
File: drivers.kicad_sch	
Title: LUM-PAR-4c drivers (4x TPS92515HV CC buck + shunt FET)	
Size: A2	Date: 2026-08-07
KiCad E.D.A. 10.0.3	Rev: 1 Id: 3/6

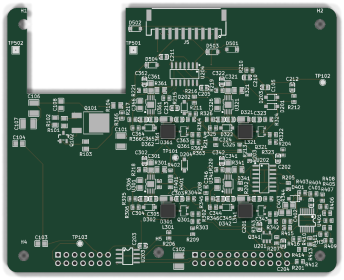

R403 10k 0603 1% C25804
R402 10k 0603 1% C25804
R405 2.7k 0603 1% C13167
R409 100k 0603 1% C25803
GND
+3V3
C401 100nF 50V X7R 0603 C14663
U401 LM339LVPWR quad open-drain comparator TSSOP-14 U42856330
R401 10k B3950 +/-1% NTC 0402
R406 270 0603 1%
R411 100k 0603 1% C25803
R407 150R 0603 1% C22808
R408 1.2k 0603 1% C22765
R410 10k 0603 1% C25804
C402 4.7uF 25V X7R 0805 C354282
C403 4.7uF 25V X7R 0805 C354282
R412 100k 0603 1% C25803
R404 10k 0603 1% C25804
R413 1.6k 0603 1% C22847
R414 100k 0603 1% C25803
D401 1N4148W switching diode, SOD-123 C81598
R415 10k 0603 1% C25804
HYSTERESIS RETURNS TO CMP_LED / CMP_BRD, NOT to the divider nodes. On the divider the emitter-OPEN channel gets NET NEGATIVE feedback and oscillates between Rntc 150k and 200k — a fracturing crimp passes through that window. R410 = 10k is the open channel's ONLY positive path.
U401 INPUT PROTECTION (R413/R416 + D401): LM339LV inputs are abs-max 6 V and are clamped to GND ONLY — there is no ESD clamp to V+. /NTC_LED shares the J5 harness with LED anodes that reach the SMF15A 16.7–24.4 V clamp region. Series R alone does NOTHING without D401. DO NOT REMOVE EITHER, and keep D401's band toward +3V3.
al-ee
Sheet: /thermal/
File: thermal_kicad_sch
Title: LUN-PAR-A: thermal (NTC dividers, window comparator, FAULT)
Size: A3 | Date: 2026-08-07 | Rev: 1
KiCad 8.0.4, 10.0.3 | Id: 2/6


S10B-PH-SM4-TB JST PH 10-way SMD header, 2.0 mm
J5
LED0
GND
LED1
GND
LED2
GND
LED3
GND
GND
GND
CONN_10
GND
GND
D501
LED0
GND
SMF15A 15V unidirectional TVS, SOD-123
D502
LED1
GND
SMF15A 15V unidirectional TVS, SOD-123
D503
LED2
GND
SMF15A 15V unidirectional TVS, SOD-123
D504
LED3
GND
SMF15A 15V unidirectional TVS, SOD-123
LED0
LED1
LED2
LED3
TP501
thermocouple pad, bare Cu
GND
ICD s9 BENCH HAZARD silk — floating at PoE potential
TP502
thermocouple pad, bare Cu
GND
ICD s9 BENCH HAZARD silk — floating at PoE potential
TP501/TP502 CARRY THE ICD s9 BENCH-HAZARD SILKSCREEN. These are BARE-COPPER pads on GND, and this board's GND floats at PoE potential: an earthed scope probe or a non-isolated USB-UART adapter ties it to earth and BREAKS PD SIGNATURE DETECTION outright (detection is a few hundred uA), on top of the shock and damage risk. Probe only through an isolated instrument. Silk text is a P6/P7 place_edit add_text op.
J5 IS SIDE ENTRY AND ITS LAND IS ASYMMETRIC: the two solder tabs sit 0.2 mm beyond the wafer's MATING end face, the 10 circuit pads 5.5–9.0 mm behind it. P6 must paint that end face OUT across the top edge and must NOT mirror the footprint — mirrored puts the opening on the wrong side of the edge. Check it in the MATED view, as ICD s7.2 requires for J3/J4.
PIN 9 IS A DEDICATED NTC SENSE RETURN. It is GND in the netlist and there is no way to say more there, so P6/P7 owns it: pin 9's copper must reach GND at the analogue reference point near U401, NOT be stitched into the plane at the connector (blocks_md B6). No gate enforces this.
al-ee
Sheet: /led_lf/
File: led_lf.kicad_sch
Title: LUN-PAR-A: led_lf — J5 LED harness, TVS clamps, TC pads (B6)
Size: A3
Date: 2026-08-07
Rev: 1
KiCad 8.0.4, 10.0.3
Id: 2/6

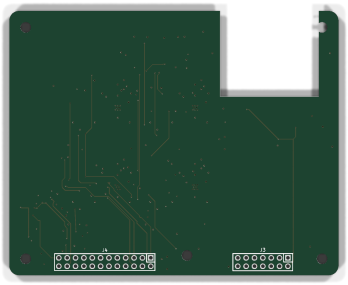
6 Layout

Gate `place` (phase P6): **pass** after 1 attempt(s); last 2026-08-07T20:05:19, failing 0 of 0.

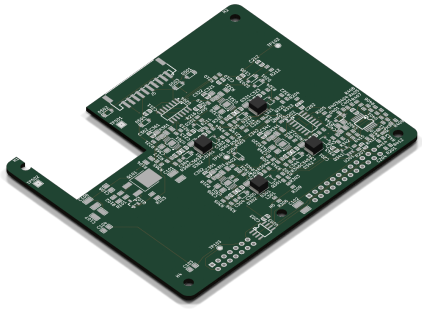
Gate `drc_routed` (phase P8): **pass** after 3 attempt(s); last 2026-08-08T01:02:43, failing 0 of 0.



`reports/renders/lumina-par_top.png`



`reports/renders/lumina-par_bottom.png`



`reports/renders/lumina-par_iso.png`

7 Verification

Gate `verify` (phase P8): **pass** after 4 attempt(s); last 2026-08-08T12:30:46, failing 0 of 60. *reports/verify_all.json not available.*

Design review of record

LUM-PAR-A board review (P8, adversarial)

Reviewer: fresh-context P8 board reviewer. No part of this board was placed or routed by me. Every number below is measured from `kicad/lumina-par.kicad_pcb` or from an artifact named inline; renders are the ones I generated at `reports/renders/lumina-par_{top,bottom,iso}.png`.

Board-local coordinates are ICD frame (origin = board top-left, x right, y down). PCB-absolute = board-local + (54.9375, 57.1925). Both are given where it matters; the JSON carries PCB-absolute so `fix_dispatch` can find things.

Gate state I am reviewing against, not from: erc 0/0, place 0/0, drc_routed 0/0, verify PASS (0 failing, 3 waived, 57 warnings). Nothing below is contradicted by a gate - that is the point. Where a gate is structurally incapable of seeing a defect, I say which gate and why.

5 errors, 15 warnings, 4 waiver recommendations.

—

Errors

E1. H1's mounting hole is cut open by the RJ45 notch - the corner has no standoff H1 sits at board-local (5.000, 5.000) with a 3.2 mm drill. The RJ45 relief's left wall is an `Edge.Cuts` line at x = 6.000 (absolute 60.9375) running the full 26 mm of the notch. Hole centre to that edge is **1.000 mm** against a 1.600 mm hole radius, so:

- **1.04 mm2 of the 8.04 mm2 hole lies outside the board** (13 %). The hole is not a hole: it opens into the notch as a **2.50 mm wide slot**.
- An M3 screw at H1 has no material on the notch side. An 11 mm F-F standoff body (~5 mm across) overhangs by 4.95 mm2, an M3 pan head (5.5 mm) by 6.50 mm2, a plain washer (6.0 mm) by 8.25 mm2.
- What is left of the 6 mm finger between the board edge and the notch is **1.4 mm of FR4 at the narrowest**, with a routed slot through it - a crack initiator at exactly the corner a technician grips while unmating.

Visible in `lumina-par_iso.png` (near-left corner, marked H1: compare its scalloped opening against the closed circle at H4) and in `lumina-par_top.png` at the top-left of the notch.

Root cause is in ICD-01, not in this board. s7.1 puts the four corner holes at a 5 mm inset; s7.6 starts the RJ45 relief at x = 6 and says "the outline rectangle, corner radius and 5-hole pattern are **unchanged**". A 3.2 mm hole centred at x = 5 needs material out to x = 6.6. The two clauses cannot both hold, and **every LUMINA daughter inherits the collision** - the carrier does not, because it has no notch. Per ICD s10 this is a blocking issue against LUM-CAR-A, not a local variant.

Why no gate sees it. KiCad's `edge_clearance` constraint applies to copper items; H1 is `MountingHole_3.2mm_M3`, an NPTH pad with no copper layer, so DRC has nothing to test. `check_*` has no mechanical-outline check at all.

Two candidate fixes, both needing the carrier owner: move H1 to (3.0, 5.0) (1.4 mm to the edge, 1.4 mm to the notch - still poor), or start the notch at x = 8 and shorten it to 28 mm. Neither is this board's call.

E2. The ICD s9 bench-hazard silkscreen is claimed in the design doc and does not exist on the board `reports/design_doc/lumina-par-design-doc.tex` states, twice, that the marking is fitted: *"TP103 carries the ICD s9 bench-hazard silkscreen"* (test-point table) and *"TP501, TP502 … ICD s9 bench-hazard silkscreen"*. `requirements.md` s2.5 makes it mandatory: *"Any test point fitted must carry the ICD s9 bench-hazard warning."*

The board carries **no silkscreen text of any kind other than component reference designators**. There are zero `gr_text` items on the board, and every `fp_text` that is not a refdes is on `F.Fab` / `B.Fab`, which is not fabricated. Confirmed visually: no legend anywhere in `lumina-par_top.png` or `lumina-par_bottom.png`.

TP103 is a bare 1.5 mm pad on `+48V_SW` at board-local (22.06, 71.31) - 57 V worst case, on a fixture that floats at PoE potential with no chassis earth. ICD s9's published consequence of clipping an earthed probe or a non-isolated USB-UART there is not merely shock and damage: it **breaks PD signature detection outright**, because detection currents are a few hundred microamps. TP501/TP502 are bare thermocouple pads on GND - the same floating return.

A safety marking that the design record asserts and the fabricated board will not have is worse than one that was never promised: the next person reads the design doc, believes the board is marked, and does not add it.

E3. U341 has 6 thermal vias where its three identical siblings have 8, and the gate cannot see the difference Measured inside each declared thermal pad's polygon (+0.05 mm halo):

ref	declared ‘min_vias‘	vias found
U301	9	8
U321	9	8
U341	9	**6**
U361	9	8
D301 / D321 / D341 / D361	4 each	**0 each**

The 8-not-9 result on three drivers is the documented geometric maximum at the 0.5 mm hole-to-hole floor and is fine. **U341 at 6 is not** - it is a 25 % conduction deficit against three parts that are the same footprint, the same power (0.15 W declared) and the same channel topology, with no recorded reason. On a fixture whose whole purpose is colour consistency (PAR-REQ-06, carried at reduced confidence since AMD-01), a thermally asymmetric channel is a chromatically asymmetric channel.

Why no gate sees it. `check_thermal` only emits its `thermal_vias` violation when `need_vias` is true - that is, when the copper-area model alone cannot reach `dt_c`. At 0.15 W and 0.14 W it can, so `need_vias` is False and **all eight declared ‘min_vias‘ floors are never evaluated**. The board declares eight thermal-via requirements and the P8 suite tests none of them. `check_thermal` reports `pass` with 0 violations, truthfully, about something else.

The four catch diodes at zero vias are carried separately as W4 - their declared floor of 4 may not even be geometrically achievable in a 1.05 x 1.5 mm land, which is itself a constraints defect rather than a layout one.

E4. The J3/J4 pin-1 marker is unprintable, is the wrong shape, and points at the wrong row Three separate defects in one 0.12 mm feature.

(a) It will not print. The marker on both connectors is an `fp_circle` with `center` and `end` 0.03 mm apart - radius 0.03 mm - stroked at **0.06 mm** with `fill no`. Outside diameter 0.12 mm. JLC's silkscreen minimum line width is ˜0.153 mm. This is roughly a third of the minimum feature and will be dropped or smeared by the screen. J3's marker is at board-local (13.81, 73.27), J4's at (55.66, 73.33), both on `B.SilkS`.

This is systemic, not local: **89 silk graphics on this board are stroked at 0.06 mm and 43 at 0.12 mm**, all below the fab floor. The DRU does carry `aiee_silk_width_floor` at 0.150 mm, but its condition is `A.Type == 'text' && ...`, so it constrains only text objects. No rule anywhere tests graphic silk width, which is why drc_routed is 0/0.

(b) It is a circle, not a triangle. ICD s3 ("Position 1 is silkscreen-marked with a triangle on **both** boards") and s7.4 mechanism 5 ("A pin-1 triangle at position 1 of both blocks on **both** boards") both specify a triangle.

(c) It marks the wrong row. After the reverse-mount flip, footprint pad 1 is at board-local (15.379, 74.730) on J3 and (57.029, 74.730) on J4. The carrier's position 1 - per ICD s7.2 rev A7, the re-issued as-built table - is at (15.380, 77.270) and (57.030, 77.270): **the other row**. So the daughter's pin-1 mark sits 2.54 mm from the carrier's and identifies the contact that mates with the carrier's position **2**.

ICD s7.4 lists five anti-mis-mate mechanisms and silkscreen is the fifth. Here it does not merely fail to help - a technician who checks that the two pin-1 marks line up will read a one-row offset and conclude the pair is mis-seated when it is correct. Both connector bodies also sit over their own

silk (the sockets are 8.5 mm and cover the outline), so the corner mark is the only part of it visible after assembly, and it is the part that is wrong.

E5. The board's J3/J4 pad numbering is the mirror of the ICD map that requirements.md prints as authoritative **The mating geometry is correct - I verified all 38 positions and found no electrical error.** Every net on the daughter lands on the carrier pin carrying the same net under ICD s7.2 rev A7:

- J3: `+48V_SW` on daughter pads 2/4/6 at board-local y = 77.269, which is where the carrier's positions 1/3/5 are; GND opposite at y = 74.730 against the carrier's 2/4/6; `+12V` on 10/12, `+3V3` on 11/13, GND on 7/8/9/14 - all matching the carrier pin at the same (x, y).
- J4: PWM0 on daughter pad 2 (carrier position 1), PWM1 on pad 1, PWM2 on 6, PWM3 on 5, `I2C_SCL` on 17 (carrier 18), `I2C_SDA` on 20 (carrier 19), `ADC0` on 19 (carrier 20), `ADC1` on 22 (carrier 21), `ID_ADC` on 21 (carrier 22), `ENABLE` on 24 (carrier 23), `FAULT` on 23 (carrier 24), and every carrier PWM4-7 / DSPI position landing on an unconnected daughter pad.

The problem is that the daughter's pad numbering is now offset by one row from ICD position numbering - board pad n carries ICD position n+1 for odd n and n-1 for even n, on both connectors - and nothing in the project's documents says so:

- `requirements.md` s2.1 still prints *"Pin map (ICD s3.1), authoritative, do not re-derive: 1/3/5 `+48V_SW`; 2/4/6/7/8/10/13 `GND`; 9/11 `+12V`; 12/14 `+3V3`"*. On this board J3.1 is GND and J3.2 is `+48V_SW`.
- s2.2 likewise prints *"'PWM0..7' at positions 1,2,5,6,7,8,11,12"*. On this board J4.1 is PWM1.

Anyone who brings this board up with a meter against `requirements.md` reads a 57 V rail on the wrong pin. This is the same class of defect as the row-swap that was caught this session, one artifact downstream - and it is the artifact a bench technician actually opens. It needs a daughter-side pad table stating that daughter pad n mates carrier position n-/+1, published in `requirements.md` s2.1/s2.2 and the design doc, before this board is handled powered.

—

Warnings

W1. In1.Cu is not the solid GND plane the whole return-path argument rests on `architecture/constraints.json` justifies leaving GND out of `power[]` on the grounds that *"the requirement is structural instead (solid In1, per-channel harness returns, 7 connector GND pins ...)"*. Measured, In1.Cu carries **226 mm of foreign track and 106 foreign vias**:

net on In1.Cu	length	board-local extent
'+3V3'	87.2 mm	(40.7, 25.4) -> (91.6, 68.9)
'/NTC_LED'	58.9 mm	(45.7, 13.2) -> (90.3, 73.4)
'/LED0_A'	38.8 mm @ 0.4 mm	(52.5, 17.2) -> (62.0, 51.3)
'/EN_OK'	14.2 mm	(34.4, 30.6) -> (46.3, 33.7)
'/thermal/CMP_LED'	14.1 mm	(85.7, 58.6) -> (93.3, 69.5)
'/ADC0'	12.6 mm	(51.5, 55.0) -> (79.9, 74.7)

In2.Cu (+12V) carries a further 166 mm and 195 foreign vias, including `/SHUNT0` 9.1 mm and `/LED0_A` 7.9 mm.

The `/LED0_A` run is a 38.8 mm slot straight down the middle of the driver band - the P8 fix that took `check_current` from 36 to 0 bought its IPC width by cutting the reference plane. `+3V3` and `/NTC_LED` between them thread the plane over most of the board's width and height.

The plane is not fragmented, and I want that on the record separately: the In1 GND fill is 5833 mm2 in 5 pieces - one of 5826 mm2 plus four slivers of 0.75-2.70 mm2 between J3's pads at board-local x 17-24, each touching 1-3 THT pads and therefore connected through the barrels. Nothing is orphaned. The defect is slot topology, not islands.

The only check that would see this is `check_return_path`, and it is scoped to the four nets declared in `constraints.high_speed`. Six nets cut the plane; none of the six is checked as a victim.

W2. The right-hand 12 mm strip has no reference plane at all - and the independent over-temperature comparator lives in it Both pours stop at board-local **x = 88 over the full 80 mm height** (zone outlines (54.937, 57.192)-(142.937, 137.192) on In1 and In2). ICD s7.6 only reserves y 25-55 in that column. The removal outside 25-55 is a choice, not a constraint.

U401 (LM339LV, TSSOP-14) spans board-local x 89.1-93.0, y 61.5-67.1 - **6.5 mm clear of the reserved column and entirely off both planes** - with R403-R416, C401-C403 and D401 around it. Measured in the (84, 54)-(100, 78) box: 50.0 mm2 of In1 GND (all of it at x < 88), 6.1 mm2 of F.Cu GND, 0.9 mm2 of B.Cu GND and **4 GND vias**. Over the whole x > 88 strip (956 mm2 of board) there is 2.39 mm2 of F.Cu GND and 0.83 mm2 of In1 GND.

This block is the board's hardware backstop: `/FAULT` is driven by U401.1/2/13/14 wire-OR'd to the carrier. `architecture/p4-wiring-notes.md` s3 records that the LM339LV has **no internal hysteresis**, so all of it is external (R405-R412). A comparator bank with external-only hysteresis, no ground reference and four switching converters 25 mm away is how a few millivolts of offset becomes chatter on the fixture's fault line. `check_return_path` never looks here.

The plane can legally be restored at x 88-100 for y < 23 and y > 57, which covers the whole analog block while leaving the ICD column void intact.

W3. PWM0-3 return through the +12V plane, not GND - 11 of the 13 return-path warnings say so and the class matters more than the geometry Stackup from the board: F.Cu | 0.2444 | In1(GND) | 1.0650 | In2(+12V) | 0.2444 | B.Cu. **Every B.Cu trace on this board references In2 = +12V**, 0.244 mm away, not GND at 1.31 mm.

net	total	on B.Cu
‘/control/PWM0‘	38.6 mm	29.6 mm
‘/control/PWM1‘	50.3 mm	28.8 mm
‘/control/PWM2‘	52.6 mm	38.3 mm
‘/control/PWM3‘	45.0 mm	36.2 mm
‘/SHUNT0‘	72.0 mm	48.3 mm
‘/SHUNT2‘	42.8 mm	23.3 mm
‘/SHUNT1‘	55.4 mm	19.4 mm

`constraints.high_speed` declares `"reference": "GND"` with `t_rise_ns: 2.0` for all four PWMs. Eleven of the thirteen `check_return_path` warnings carry `waiver_class: "cross_net_reference"`, `reference_declared: "GND"`, `reference_net: "+12V"` - the checker is telling you it measured a different plane from the one declared, and then measuring corridor voids in it.

Individually the deficits are 0.13-1.75 mm2 and are not worth a fix. **The class is:** roughly 133 mm of 2 ns-edge PWM and 91 mm of shunt-gate drive inject their return current into the rail that feeds all four LED channels. `+12V` is well bypassed to GND (2 x 22 uF 1210 plus 4 x 4.7 uF at the drivers) so it is a serviceable AC ground - but that is an argument that should be written down and waived, not left as eleven silent warnings whose text says the reference is wrong.

Recommendation: either declare `+12V` as the reference for the B.Cu segments and re-run, or move PWM0-3 to F.Cu where In1 GND is 0.244 mm away. Do not leave the declaration and the copper disagreeing.

W4. Four declared thermal sites have zero of their required vias - and the requirement may be impossible as written D301/D321/D341/D361 each declare `min_vias: 4` in `constraints.json`; each has **0**. Their GND land is 1.05 x 1.5 mm. Four 0.45 mm vias at the 0.5 mm hole-to-hole floor do not fit in 1.6 mm2; two might. So this is a constraints defect as much as a layout one - but it is a declared requirement that is unmet and unmeasured (see E3 for why `check_thermal` is silent). The declared 0.14 W is the shunted-idle case, which is the *worst* case at unchanged channel current, so it is a steady-state number on a continuous-duty board.

W5. 42 refdes name the wrong part - that is a mis-build mechanism on this board, not cosmetics All 42 `check_silk` warnings are `silk_misattributed`: **27 % of the 155 reference designators sit closer to a neighbouring part than to their own.** The worst are effectively touching:

- R410 - **0.00 mm** from U401
- C363 - 0.06 mm from L361
- C204 - 0.09 mm from J4
- R342 - 0.12 mm from U201
- R416 - 0.15 mm from R414
- R411 0.23 mm from R412, R412 0.19 mm from H3, R344 0.23 mm from R205, H3 0.29 mm from R411, R325 0.41 mm from R214 ...

Ordinarily this is a warning you take on the chin. Not here. **30 parts on this board are DNP** (C106-C108, C210-C213, C303/C323/C343/C363, D201-D204, Q101, Q102, R101-R104, R210-R213, R304/R324/R344/R364, U204) and - per the already logged BOM item - they will ship. A build operator resolving "which of these two 0603s is C363 (DNP) and which is L361 (fit)" from a legend that is 0.06 mm from the wrong part will get it wrong, and the shunt-FET snubbers and the converter-idle one-shot are precisely the parts that must **not** be fitted on branch A. `check_silk` even names the scripted fix (`place_edit.py move_text`). This is real risk; I do not recommend waiving it.

W6. `check_pdn`'s "+3V3 has no bulk reservoir" is a false alarm, and the reason it fired is worse than the warning The rail does have bulk: **C103 and C104 are 10 uF 16 V X7R 0805 on `+3V3`/GND**. The warning listed only C201, C202, C203, C205, C401.

`check_pdn` inventories only capacitors listed in `kicad/decoupling.json`. Absent from that file: **C101, C102, C103, C104, C105, C402, C403** - both `+3V3` bulk parts, both 22 uF `+12V` bulk parts, and both ADC filters. So the PDN check is blind to 7 of the board's capacitors, and `check_decoupling` (which reported `pass`) is working from the same partial list.

Verdict on the warning itself: **waive** - the hardware claim is wrong. The finding is the metadata gap, which silently narrows two P8 checks.

W7. Both bulk reservoirs are parked in the vacant left third, far from entry and far from load

```
| cap | board-local | to its J3 entry pad | to nearest load |
|---|---|---|---|
| C103 (10 uF, +3V3) | (10.3, 71.3) | 18.1 mm | 26.6 mm (U203); U202 70.6, U401
81.1 |
| C104 (10 uF, +3V3) | (3.6, 41.1) | 41.7 mm | 47.5 mm (U203) |
| C101 (22 uF, +12V) | (34.6, 40.3) | 38.0 mm | 14.2 mm (U361) |
| C102 (22 uF, +12V) | (52.1, 76.6) | 26.5 mm | 24.8 mm (U301) |
```

+3V3 arrives on two pins of a 2.54 mm connector - tens of nH of loop - and its only reservoir is 18 mm away in the opposite corner from every load. The four per-channel 4.7 uF input caps do the real work on +12V; C101/C102 at 26-38 mm from the connector are contributing little at the entry and little at the loads.

The cause is visible in lumina-par_top.png: the whole left third (x 0-36, y 26-80) is empty apart from the DNP branch-B front end, while the analog block is squeezed out past the plane edge at x > 88 (W2) and 42 refdes collide (W5). The annealer had a large free region and used it for the parts that least needed it.

W8. Antenna column: verified genuinely clear, nothing orphaned, two residuals Answering the specific question directly. Measured copper strictly inside the board-local rect (88, 25)-(100, 55):

```
F.Cu 0.0000 mm2 In1.Cu 0.0000 mm2
In2.Cu 0.0000 mm2 B.Cu 0.0000 mm2
```

The rule area aiee_antenna_column_no_copper is present on all four layers at absolute (142.937, 82.192)-(154.937, 112.192), which is the ICD rect translated exactly. The inner pours stop at x = 142.937 as intended. Visible as the bare right-hand column in lumina-par_top.png.

Nothing was orphaned by stopping the plane there. In1 GND fills 5833 mm2 in 5 pieces: one of 5826 mm2, and four slivers of 0.75/1.41/1.94/2.70 mm2 located at board-local x 17-24, y ˜ 75.6-76.0 - between J3's pads, nowhere near the column - each touching 1-3 through-hole pads and therefore tied through the barrels. In2 +12V is a single piece of 5859 mm2. No net lost connectivity to the truncation.

Two residuals:

- **The rule area allows 'pads' and 'footprints'** (pads allowed, footprints allowed; only tracks, vias and copperpour are not_allowed). Today the column is clean, but nothing prevents a later place_edit or route_edit from dropping a pad or a part in it, and DRC will not object.
- **Two metal features sit just outside the boundary, in the antenna's near field 11 mm above the radiator:** R212's pads at board-local x 87.06 (0.94 mm west of the wall) and TP102, a bare 1.5 mm pad at (95.81, 22.56), 2.44 mm north of it. Both compliant with s7.6 as written. Flagging because s7.6's rectangle is a minimum, not a guarantee, and Q8 kept the radio a supported control path.

W9. Reverse-mounted THT on a fully populated top side is a hand-solder-only operation and no fab note says so The mezzanine geometry itself is clean: **J3 and J4 are the only bottom-side parts on the board** (157 of 159 footprints are F.Cu), their 8.5 mm bodies leave 2.5 mm for the male insulator in the 11.0 mm gap, and there is nothing else on B.Cu to foul it. lumina-par_bottom.png shows a bare bottom side apart from the two sockets and routing.

The assembly consequence is the problem. The bodies are on the bottom and the **38 leads are soldered on the top face, which carries the entire SMD assembly**. That pair cannot go through a wave or a selective nozzle without masking the whole top side; it is manual. Nearest top-side parts to a J3/J4 pad: C204 2.96 mm, R201 3.18 mm, U203 3.27 mm, TP103 3.55 mm, R207 3.72 mm - workable with a fine conical tip, not with a standard chisel, and U203 is the EEPROM.

Nothing in the fab package flags it. Quoted as ordinary THT, this gets built wrong or gets re-quoted late.

W10. +48V_SW clears the binding 0.635 mm by 12 um - less than the etch tolerance Measured minimum outer-layer copper gap on the 48 V net:

```
| where | gap | vs ICD 0.635 mm |
|---|---|---|
| F.Cu, ‘+48V_SW‘ <-> ‘+3V3‘, board-local (15.96, 71.21) | **0.6470 mm** |
+0.012 mm |
| F.Cu, ‘+48V_SW‘ <-> GND, (14.07, 76.04) | 0.6516 mm | +0.017 mm |
| B.Cu, ‘+48V_SW‘ <-> GND, (14.06, 76.05) | 0.6611 mm | +0.026 mm |
```

All pass. All three are inside a typical +/-0.05 mm outer-layer etch tolerance of the requirement, so on a bad panel the binding creepage figure is not met. `check_creepage` demands 0.60 and passes by construction; the DRU rule `aiee_hv_57v_48V_SW` at 0.635 mm is satisfied by 12 um at the worst point.

ICD s5.2 names the lever and it has not been pulled: *”if P6/P7 needs more margin, shrink the annulus on the 48 V pads to 1.60 mm before moving anything else.”* J3’s pads are still 1.70 mm. Dropping the three `+48V_SW` pads to 1.60 mm buys 0.05 mm at zero routing cost.

W11. The V48_B 0.590 mm exclusion is correctly scoped - evidence added, not re-opened Confirming the DRU’s own note rather than re-deriving it. Per-neighbour distance from `/power/V48_B` on F.Cu:

```
0.5900 mm -> /power/Q101_G (C108’s own 0805 land)
0.6626 mm -> GND
0.6637 mm -> /EN_OK
```

The `B.NetName != ’/power/Q101_G’` carve-out in `aiee_hv_57v_V48_B` hides exactly one pair and nothing else on that net drops below 0.635 mm. Q101 is DNP on branch A so V48_B is never energised on the built board. Carry it on the branch-B populate checklist, not on this build’s defect list.

W12. The recovery-header choice was made by omission ICD s7.6 offers a binary: keep board-local (76, 0)-(98, 20) clear enough that a 6-way jumper lead can be attached with the daughter fitted, **or** accept that the daughter must be removed to recover firmware. The board presents solid FR4 over the whole rect (only H2 is inside it) at an 11.0 mm stack height. A 6-way jumper on 2.54 mm sockets plus wire bend does not fit in 11 mm, so the board has chosen option two.

`requirements.md` s5.2, `architecture/decisions.md` and the design doc all quote the ICD sentence verbatim and none of them records a verdict. The accepted cost is: unbolt five 11 mm standoffs and unmate a 38-position pair every time the carrier’s firmware needs recovery, on 6-8 fixtures. That is a decision someone should sign, not inherit.

W13. The design doc’s surviving rationale for J5 says it is on B.Cu; it is on F.Cu The board is right and the entry direction is correct - I verified it. J5 is `S10B-PH-SM4-TB`, side entry, on **F.Cu** at board-local (54.416, 4.967), rotation 0, not mirrored. The land is asymmetric front-to-back exactly as it should be: the ten 1.0 x 3.5 mm signal pads sit inboard at y = 7.738 and the two 1.5 x 3.4 mm board-lock pads sit outboard at y = 2.197, with the silk body front face drawn at y = 0.167. The wafer opening therefore faces **-y, out across the top edge**. It clears the RJ45 notch (J5 spans x 42.3-66.6 against the notch’s 6-36) and the recovery keepout (76-98).

But the P4 close-out in `lumina-par-design-doc.tex` reads: *”...and J5 is on B.Cu inside the ICD’s 11.0 mm mezzanine gap where a top-entry PHR-10 housing plus bend radius would foul the carrier.”* J5 is not on B.Cu. A 10-way PH in the mezzanine gap would be a defect; the board avoided it. The stale sentence is the record a future reviewer will use to decide not to re-check, and it happens to describe the failure mode it claims to have avoided.

W14. No fiducials, with 0.5 mm pitch on the board Zero fiducial footprints. The board carries 4 x MSOP-10 with exposed pad at 0.50 mm pitch (the TPS92515HV drivers) and 1 x TSSOP-14 at 0.65 mm (U401). JLCPCB does not require fiducials and registers from the board edge, so

this is not a blocker - but 0.50 mm pitch is where local fiducials start to earn their place, and the board has 956 mm2 of free space in the right-hand strip and a whole empty left third to put them in. Low priority; decide once rather than discovering it at first-article.

W15. `verify_all`'s constraints-drift warning is expected and should stay expected `architecture/constraints.json` and `kicad/constraints.json` differ only in the documented coordinate frame: every `placement.keepouts` rect and every `planes[].region` in the kicad copy is the architecture copy translated by (+54.937, +57.192), plus a `_coordinate_frame` key explaining it. I diffed both files in full; there is no other difference. **Justified waiver** - but it will fire on every P8 run forever, and the remediation text ("reconcile or delete the stale twin") is wrong for this board, because the board-local copy is the source of truth per its own header.

—

Triage summary of the 57 verify warnings

source	n	verdict
‘check_return_path‘, In2.Cu / ‘+12V‘ reference, ‘waiver_class: cross_net_reference‘	11	**Waive individually** (0.13–1.75 mm2 each), **escalate as a class** -> W3
‘check_return_path‘, In1.Cu / GND, PWM1 on F.Cu, 0.14 and 0.29 mm2 at board-local (72.5, 44.6) and (72.1, 45.0)	2	**Justified waiver** - antipad-scale voids in an otherwise continuous plane
‘check_silk‘ ‘silk_misattributed‘	42	**Real risk, do not waive** -> W5
‘check_pdn‘ ‘pdn_no_bulk‘ on ‘+3V3‘	1	**False alarm, waive the claim** - C103/C104 are 10 uF; the finding is the ‘decoupling.json‘ gap -> W6
‘verify_all‘ ‘constraints_drift‘	1	**Justified waiver**, permanent by design -> W15
total	**57**	15 waived, 42 escalated

Plus the 3 waived `check_silk` errors (TP101/102/103 rings): already logged as a checker bug, not re-opened. I add one datum only - those three test points are also the ones E2 says should be carrying a hazard legend and are not.

—

Waiver recommendations for checkpoint 4

1. **11 x ‘check_return_path‘ cross-net-reference** - waive the individual geometries; require the class decision in W3 to be recorded (declare `+12V` as the B.Cu reference, or move PWM0-3 to F.Cu). 2. **2 x ‘check_return_path‘ In1/GND corridor voids** - waive outright. 3. **1 x ‘check_pdn‘ ‘pdn_no_bulk‘** - waive the hardware claim; open a `decoupling.json` completeness item instead. 4. **1 x ‘verify_all‘ ‘constraints_drift‘** - waive permanently with a note that the twin is intentional and directional.

Not recommended for waiver: the 42 `silk_misattributed` (W5), and all five errors.

—

What I could not judge

- **Component heights.** The board has no 3D height data I can trust for anything but J3/J4 (8.5 mm from the ICD). L301/L321/L341/L361 are `L_Cenker_CKCS4030` - a 4.0 x 4.0 mm part whose height I did not confirm. They are on the top side so the 11 mm mezzanine gap is not at risk, but the enclosure lid clearance and the J5 harness bend radius over the top edge are outside what a render can settle.

- **Whether U301 is genuinely the hottest stage.** RT401 is 2.10 mm from U301's pads, sitting over In1 GND, and 19.9-29.0 mm from the other three drivers, so it is a channel-0 sensor, not a board-average one. All four drivers are declared at an identical 0.15 W in `constraints.json`, so the declaration cannot distinguish them; `power_tree.md` gives the red string a different duty from G/B/W, which should make one stage hotter than the rest. Which channel is red, and whether it is channel 0, is a schematic question I did not chase.
- **Pin-1 and polarity marks on the SMD parts after assembly.** I confirmed the connector markers (E4) but did not audit all 157 top-side footprints for post-assembly visibility of their own polarity marks; with 42 refdes already misplaced (W5) it is worth a dedicated pass.
- **The enclosure.** Wall conduction is the load-bearing assumption in ICD s7.7.3 and none of it is on this board.

8 DFM and Fabrication

Pending — produced at P9.

9 Run Record (Appendix)

Phase digests

P0

```
# P0 Intake digest - lumina-par (LUM-DTR-PAR-A / LUM-PAR-A)

- Board:  RGBW par daughter for LUM-CAR-A. 4 (maybe 5) PWM-dimmed
constant-current LED
channels, stacked 11.0 mm above the carrier on the frozen ICD-01 connector
pair.
Continuous duty; the design problem is dimming quality and fixture-to-fixture
colour
consistency, not peak power.
- Inherited as CLOSED (05-lumina-closed-decisions.md):  D-01 (at-capable stage,
af-classified;
budget 8.6-9.3 W af / 18.7-20.0 W at to this daughter - the "~8.5 W" and "~10 W
regulated"
figures in briefs 00/03 are dead), D-02 (48 V + 12 V + 3.3 V on the connector),
D-03 (no UV).
- ICD-01 is a frozen hard input.  All interfaces are ICD-only:  J3 2x7 power,
J4 2x12 signal.
ICD s9 forbids adding an external connector of any kind.  Common footprint 100
x 80 mm,
R3.0, 5x M3, 1.6 mm, plus a mandatory 30 x 26 mm RJ45 notch at (6,0)-(36,26).
- Safety flags:  >30 V unconditional (48 V present on J3 whether tapped or not
- 0.60 mm outer
clearance, 100 V caps, 0805+ resistors across the 48 V domain); whole fixture
floats at PoE
potential including LED wiring and any heatsink; >3 A conditional on
per-channel LED current
that P1/P2 must publish.
- Riskiest unknowns, in order:  (1) af-vs-at sizing - +12V tops out at 15.0 W
so at-sizing forces
a >=60 V stage on THIS board, the exact thing D-02's 12 V rail existed to
avoid; (2) continuous
thermal in a non-conductive enclosure at 56-69 C internal air; (3) PAR-REQ-01's
"5-10 % of full
output" is ambiguous and changes the required driver settling time by ~40x.
```

```
- Hard derived requirement recorded:  the board must be electrically incapable
of exceeding the
carrier's fault ceilings (12 V OCP 2.0 A, 48 V eFuse 1.0 A latch-off) with
every PWM stuck at
100 %, because the PAR-REQ-11 clamp is firmware and firmware hangs.
- Pipeline risk logged for P5:  `board_init.py` has no notch/cutout option and
`kc.py` has no
outline editor, so the mandatory RJ45 relief needs a direct Edge.Cuts edit
after board_init;
and `--mounting-holes` places holes at margin/2, so the ICD's 5 mm inset needs
`--margin 10`.
- Artifact:  `requirements.md` (9 sections, 16 open questions, all with
recommendations).
```

P1

```
# P1 Research digest - lumina-par

Roster:  2x component-scout (LED driver, RGBW emitter), 1x reference-design
(continuous thermal + protection chain), 1x interface-spec (PWM dimming /
flicker / perceptual resolution).  No power-architect:  the rail tree is fixed
by
ICD-01 s6.2 and the only open variable was which rail feeds the LED stage,
which
became conflict C1 for the architect.

- **Thermal is the binding constraint, not the power budget.** A sealed
120x100x60 mm non-metallic box is 3.6-4.3 K/W internal-air-to-room (Hoffman
~4.6 W/m2K, cross-checked vs Rittal k=5.5).  Nothing on the PCB changes that
number.  Sealed caps total box heat at ~5.0 W for 45 C internal air in a 25 C
room; minus the carrier's 2.4 W that leaves ~2.6 W of daughter heat = ~3-4 W of
LED electrical power, **under half the af envelope**.
- **ICD s7.6's internal-air figures do not hold.** Its 69 C (at) is internally
inconsistent with its own 56 C (af):  convection is near-linear in dT at this
scale, so ~2x the heat must give ~2x the rise (31 K -> ~60 K -> ~85 C), and
the
independent calculation gives 89-115 C. Optimistic by 20-46 K. Raised as a
blocking issue against LUM-CAR-A under ICD s10 (OPEN-1).
- **PAR-REQ-01 at 5 % perceived fails at 13-bit by 8.9x** and needs ~17 bits;
ESP32-S3 LEDC caps at 14.  Two independent walls:  resolution (17 bits) and
driver settling (T <= 141 ns).  New fact, verified from ESP-IDF source:  LEDC
does
NOT rigidly couple frequency to resolution - "80 MHz / 2^n" is a maximum, so
14-bit is available well below 4.88 kHz.  Widens the trade space.
- **Flicker is over-specified.** IEEE 1789-2015 RP2 removes all
modulation-depth
restriction above 3 kHz; 9.766 kHz sits 3.3x above it.  The 10 kHz-over-1.2 kHz
camera call is confirmed (8.1x lower band contrast) but bounded:  clean to
120 fps, marginal at 240 fps, 960 fps unreachable at any usable LEDC setting.
- **No CC LED driver has an enable pin separate from its PWM pin.** On all five
surviving candidates the dimming pin IS the enable, so ENABLE must be gated
with
each PWM line externally.  The brief's "driver with an EN pin strongly
preferred"
is not achievable in this performance class.
- **Emitter is single-source.** The XINGLIGHT 4-in-1 is the only in-stock RGBW
power emitter above 50 mA on the whole LCSC catalogue (318-part sweep; every
Cree XLamp colour line at 0 stock), with no published Rth and no published Tj
max.  Fallback is RGB 3-in-1 + separate white at the cost of PAR-REQ-15.
```

```
- Orchestrator correction applied to ‘requirements.md‘:  s9 Q9(b) on-times were
10x
high (1.4-6.1 us -> 141-646 ns), verified by independent recompute.
Load-bearing:
it moved the strict reading of PAR-REQ-01 from "several drivers manage it" to
"no surveyed CC driver manages it by gating the converter".
```

P2

```
# P2 Architecture digest - lumina-par

Artifact:  ‘architecture/‘ (blocks.md, power_tree.md, stackup.md, sheets.md,
constraints.json, decisions.md).  This is the H1 checkpoint package.

- **6 blocks**:  rail entry (48 V landed on J3 but NOT tapped), ENABLE gating,
4x TPS92515HV-class CC buck with a shunt FET per channel, window-comparator
over-temperature protection, ID divider + I2C EEPROM, 10-way LED harness.
Emitters off-board on an aluminium MCPCB.
- **Stackup JLC04161H-3313, 4 layers**, 100.0 x 80.0 mm, 1.6 mm.  Drivers:
switch-node
containment 11 mm above a live 2.4 GHz antenna, a continuous GND reference
under
jitter-sensitive PWM, B.Cu consumed by two reverse-mounted sockets, ~120 parts
in
~45 cm2.  No controlled-impedance net exists on this board.
- **Design point 150 mA/die, 4 packages, 2S2P per channel.** Full white (the
stuck-PWM hardware backstop) = 0.718 A on +12V: 4 % under the sustained ceiling
and 2.8x under OCP. The board is electrically incapable of exceeding its
budget.
The >3 A conditional flag is CLOSED - max per-channel current is 0.30 A.
- **C1 arbitrated to +12V**, and the P1 argument for 48 V was withdrawn rather
than
overruled.  The "48 V gives 16x faster slew" claim held the inductor fixed
across a
4x change in (Vin - Vs).  With L sized for the same ripple fraction, t_r + t_f =
1/(f*k) - the rail voltage cancels out of the algebra.  Verified independently
by
the orchestrator:  4.762 us on BOTH rails at 700 kHz / k=0.30 (48 V only
redistributes 3.81/0.95 us between rise and fall).  That is 33.8x the 141 ns
budget, so no rail or inductor choice reaches PAR-REQ-01 by gating the
converter.
- **C2 resolved:  shunt-FET dimming closes the timing wall (T ~ 50-100 ns vs
141 ns,
1.4-2.8x margin); the resolution wall is NOT closable on this board** and needs
carrier firmware dithering (CR-5).  spec-dimming’s "local >=16-bit PWM IC"
class
LOSES to led-driver’s six-part survey of that exact class - no part delivers
>=150 mA/ch AND >=10 kHz AND useful resolution AND independent channels.
- **Cost ~$18-23/board** delivered (BOM ~$12.9, driver silicon 52 %) against a
$25-35 target; module + heatsink adds $8-14/fixture.
- **Sealed enclosure does not close** with the module inside (9.15 W -> 62 C
air
against a 45 C criterion).  It closes vented, or sealed with LED heat conducted
through the wall (3.21 W -> 38 C). Seven ENC-x acceptance criteria written for
whoever owns the enclosure, since it is not this board’s deliverable.
- Riskiest decision:  **emitters off-board on MCPCB (D3)** - the thermal
argument
rests on an ASSUMED 12 K/W package Rth, because the vendor publishes neither
Rth
```

```
nor Tj max, and it is the only in-stock RGBW power emitter on LCSC. At 20 K/W
even
MCPCB is marginal at af.
- Two P5 traps recorded:  fixed-outline board_init derives its origin from the
packed
component bbox (every rect in constraints.json needs a P5 translation), and the
antenna column has no automated checker on F.Cu/B.Cu.
```

P3

```
# P3 Parts + Library digest - lumina-par

Artifact:  ‘parts/parts.json‘ (47 lines), 18 datasheet extracts, ‘lib/‘ (91 sym
/
23 fp), ‘reports/fp_verify_*‘.  Ran 2026-07-28; digest written at the resume.

- Emitters 4x C22434861 RGB + 4x C48586656 white, 2S2P, 150 mA/die.
- **OPEN-8 REFUTED**:  no 140/120 deg beam mismatch - both datasheets say 140,
the
LCSC record is wrong.  Diffuser spec and its 30-45 % flux cost must be
re-derived.
- Sense 0.75 ohm (275 mA/ch); shunt FET SSM3K2615R (C22371361).
- **COFF network was MISSING, now fitted**:  12 parts, ROFF1 = 10k not TI’s
8.2k;
without it the shunt topology loses dimming linearity (TPS92515HV 8.3.4).
- CR-1 closed:  R206 = 4.7 k 1 % (supersedes sheets.md "TBD").  CR-3 adopted:
14-bit / 4.883 kHz PWM. CR-6 closed by ICD s7.7 A5.  Emitter Rth still absent.
- fp_verify:  4 violations, all adjudicated (2 off-board emitter modules,
thermal
pad; L301/D501 approved KiCad-stock land deltas).
- **To H2**:  shunt-FET switching UNVERIFIED at 3.3 V (toff 150 ns TYP > the
141 ns
pulse; both gate-path ends ungrounded -> P8 bench); compounded light budget;
power_tree.md quotes 300 mA where the board builds 275 mA.
- Tooling defects (central):  lib_pull false-pass, fp_verify blind to drill,
placelib ignores pad rotation -> hand to P6.
```

P4

```
# P4 Schematic digest - lumina-par

Artifact:  ‘kicad/lumina-par.kicad_sch‘ + 5 child sheets, built from
‘kicad/gen/*.py‘.  **erc PASS 0/0; netlist_audit exit 0.** 155 components,
76 nets, 46 BOM lines, 152 placed (30 DNP), ~$17.44/board.  H2 approved.

- ERC passed first attempt; the one fix loop went on REVIEW findings (2 of 3
left).
- **Adversarial review found 3 errors the gates cannot see**, all fixed:  COFF
ROFF2 47k->30k (off-timer could never trip while shunted - dead in 176/243
corners); ‘/NTC_LED‘ unprotected into 6.8-24.4 V harness (added R413/R415 +
D401); ‘/DRV_EN0..3‘ no pull-down, latching all drivers ON in the
+12V-before-+3V3 window (added R218-R221 at **10k, not 100k** - the enable
pin’s own 25 uA puts 100k at 2.5 V, above threshold).
- **Sheets found 3 more first**:  BOOT network allocated nowhere (12 parts, the
converter cannot switch without it); LM339LV is a 4-unit symbol that silently
floated 9 pins; hysteresis wrong in SIGN on 3 of 4 channels - a broken harness
chattered.  All four bands now positive (+8.6/+17.3/+17.3/+18.7 K).
- ADC source impedance at a broken harness 10.18k -> 9.70k, inside the ICD
ceiling for the first time (the old value bought margin with the defect).
- ‘/PWM0..3‘ are ‘/control/PWM0..3‘; root-level naming measured and rejected.
```

```
- Open to P5+:  DNP marking has no consumer (~30 parts); shunt-FET switching
and
tLEB unverified vs the 141 ns pulse (P8 bench).
```

P5 *(no digest recorded)*

P6 *(no digest recorded)*

P7 *(no digest recorded)*

P8 *(no digest recorded)*

Gate attempt history

Timestamp	Gate	Status	Failing/Total
2026-08-07T11:58:05	erc	pass	0/0
2026-08-07T13:18:16	erc	pass	0/0
2026-08-07T20:05:19	place	pass	0/0
2026-08-07T22:45:28	drc_routed	pass	0/0
2026-08-08T00:10:50	drc_routed	pass	0/0
2026-08-08T00:10:50	verify	fail	7/67
2026-08-08T01:02:43	drc_routed	pass	0/0
2026-08-08T01:02:43	verify	fail	3/60
2026-08-08T10:22:11	verify	pass	0/60
2026-08-08T11:27:06	sim	pass	0/0
2026-08-08T12:30:46	verify	pass	0/60
2026-08-14T01:20:22	dfm	fail	1/56

Issues

Id	Phase	Kinds	Severity	Status
1		coff-shunt-timer-dead, drv-en-undriven-no-3v3, hysteresis-on-sensor-node, missing-bench-hazard-note, ntc-harness-input-unprotected	error	fixed

Human checkpoints

Id	Phase	Status	When	Note
1	P2	approved	2026-07-28T20:03:50	H1 approved with 4 directed changes: (Q1) enclosure sealed + LED heat conducted through the wall; (Q2) emitter changed to RGB 3-in-1 + separate white, AGAINST the run's recommendation, chosen knowingly for a published 125 C Tj and 3.5x stock over an assumed 12 K/W Rth from a sole source; (Q3) firmware commits to PWM-domain dithering >=3-4 bits, strict gamma-2.2 reading of PAR-REQ-01 binds; (Q4) same-reel sourcing, 24C32 fitted but shipped empty, PAR-REQ-16/17 to be formally amended.
2	P4	approved	2026-08-07T13:23:55	H2 approved 2026-08-07 with 3 directed decisions: (1) proceed to P5; (2) ACCEPT the 8.6 K emitter-hot band - trip is exact at 89.9 C and only the release sits high, and widening trades against the emitter-open margin that catches a broken harness; R207 100k->10k explicitly NOT taken, its justification having been removed by the thermal rework; (3) re-derive the optical budget NOW, before P5, because the diffuser was specced against a 140/120 deg delta that OPEN-8 proved does not exist.
3	P6	skipped	2026-08-07T20:09:42	H3 is optional (default on). Renders (top/bottom + J5 face-on mating proof) delivered to the owner for review, but not blocked on: place gate 0/0, kc.py drc clean of everything but the expected unrouted nets, route probe 0.9971. Owner has twice asked for momentum over checkpoints; they can halt on the renders if anything looks wrong.
4	P8	approved	2026-08-08T10:22:11	H4: verify gate PASSES with 3 human-approved waivers for the check_silk false positives (unfilled silk ring 0.890-1.010 mm vs a 0.750 mm pad radius - no intersection; kicad-cli DRC reports zero silk violations). Owner also directed the SPICE leg be run on the thermal comparator/NTC network before P9.
5	P10	not reached		

Generation cost

Not recorded (no reports/cost.json).

10 Artifact Index

File	Size	Note
`state.json`	137,792 B	
`kicad/PCB-0005-A_lumina-par.kicad_pcb`	1,498,153 B	
`kicad/PCB-0005-A_lumina-par.kicad_sch`	20,336 B	

File	Size	Note
requirements.md	43,956 B	
architecture/blocks.md	36,027 B	
architecture/constraints.json	17,617 B	
architecture/decisions.md	63,672 B	
architecture/p4-wiring-notes.md	6,439 B	
architecture/power_tree.md	36,065 B	
architecture/sheets.md	16,917 B	
architecture/stackup.md	45,870 B	
reports/board_init.json	22,052 B	
reports/fix-silk-hazard.json	1,279 B	
reports/fix-thermal-vias.json	1,402 B	
reports/fp_courtyard_adequacy.json	2,879 B	
reports/fp_icd_annulus.json	1,956 B	
reports/fp_verify_(module)_C22434861.json	2,400 B	
reports/fp_verify_(module)_C48586656.json	2,718 B	
reports/fp_verify_D201_C81598.json	1,864 B	
reports/fp_verify_D301_C177023.json	2,244 B	
reports/fp_verify_D501_C435484.json	2,762 B	
reports/fp_verify_J3_C113344.json	2,007 B	
reports/fp_verify_J4_C92265.json	2,006 B	
reports/fp_verify_J5_C265014.json	2,383 B	
reports/fp_verify_L301_C354593.json	3,388 B	
reports/fp_verify_Q101_C115989.json	2,282 B	
reports/fp_verify_Q102_C427379.json	3,072 B	
reports/fp_verify_Q301_C22371361.json	3,247 B	
reports/fp_verify_U201_C7666.json	1,240 B	
reports/fp_verify_U202_C485072.json	1,761 B	
reports/fp_verify_U203_C7998.json	1,301 B	
reports/fp_verify_U204_C133541.json	1,778 B	
reports/fp_verify_U301_C213553.json	2,400 B	
reports/fp_verify_U401_C3658338.json	1,256 B	
reports/fp_verify_summary.json	8,616 B	
reports/gate-dfm.json	1,629 B	
reports/gate-verify-postfix.json	2,740 B	
reports/gate_drc_routed.json	382 B	
reports/lib_pin_types.json	19,471 B	
reports/lib_pull.json	5,779 B	
reports/netlist_audit.json	3,295 B	
reports/optical-budget-rederivation.md	25,523 B	
reports/planes_gen.json	1,304 B	
reports/refill-drc.json	378 B	
reports/review-board.json	23,934 B	
reports/review-board.md	29,111 B	
reports/review-schematic.json	4,062 B	
reports/review-schematic.md	18,880 B	
reports/route_auto.json	80,761 B	
reports/route_critical.json	3,909 B	
reports/schematic.pdf	491,501 B	
reports/sim.json	3,018 B	
reports/stitch_vias.json	17,007 B	
reports/top.net	237,900 B	
reports/verify-waivers.json	2,892 B	
reports/verify.json	2,760 B	